

BFS Traversal

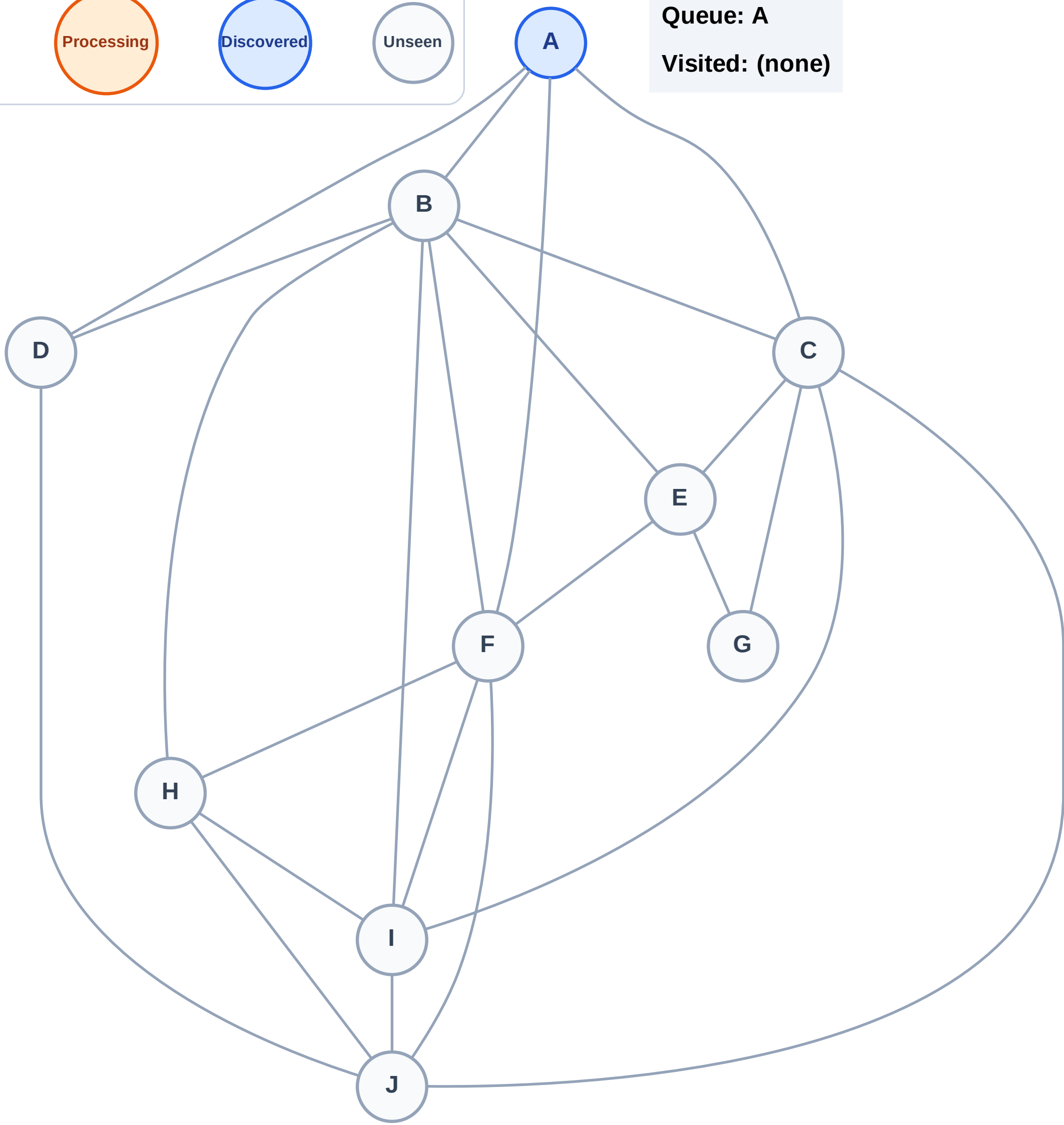
Step 1: Start at A

Legend



Queue: A

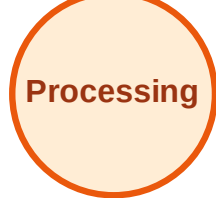
Visited: (none)



BFS Traversal

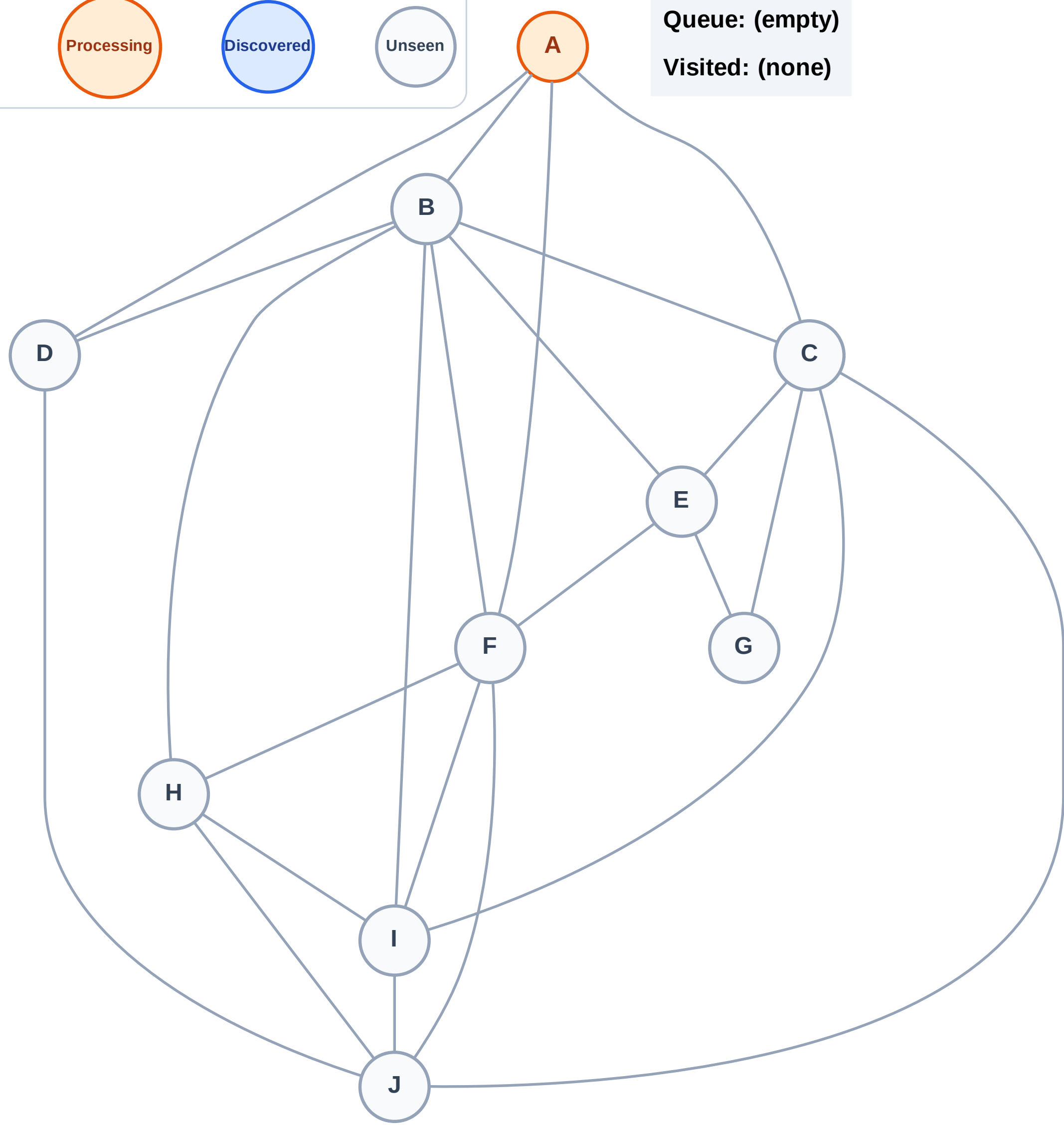
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

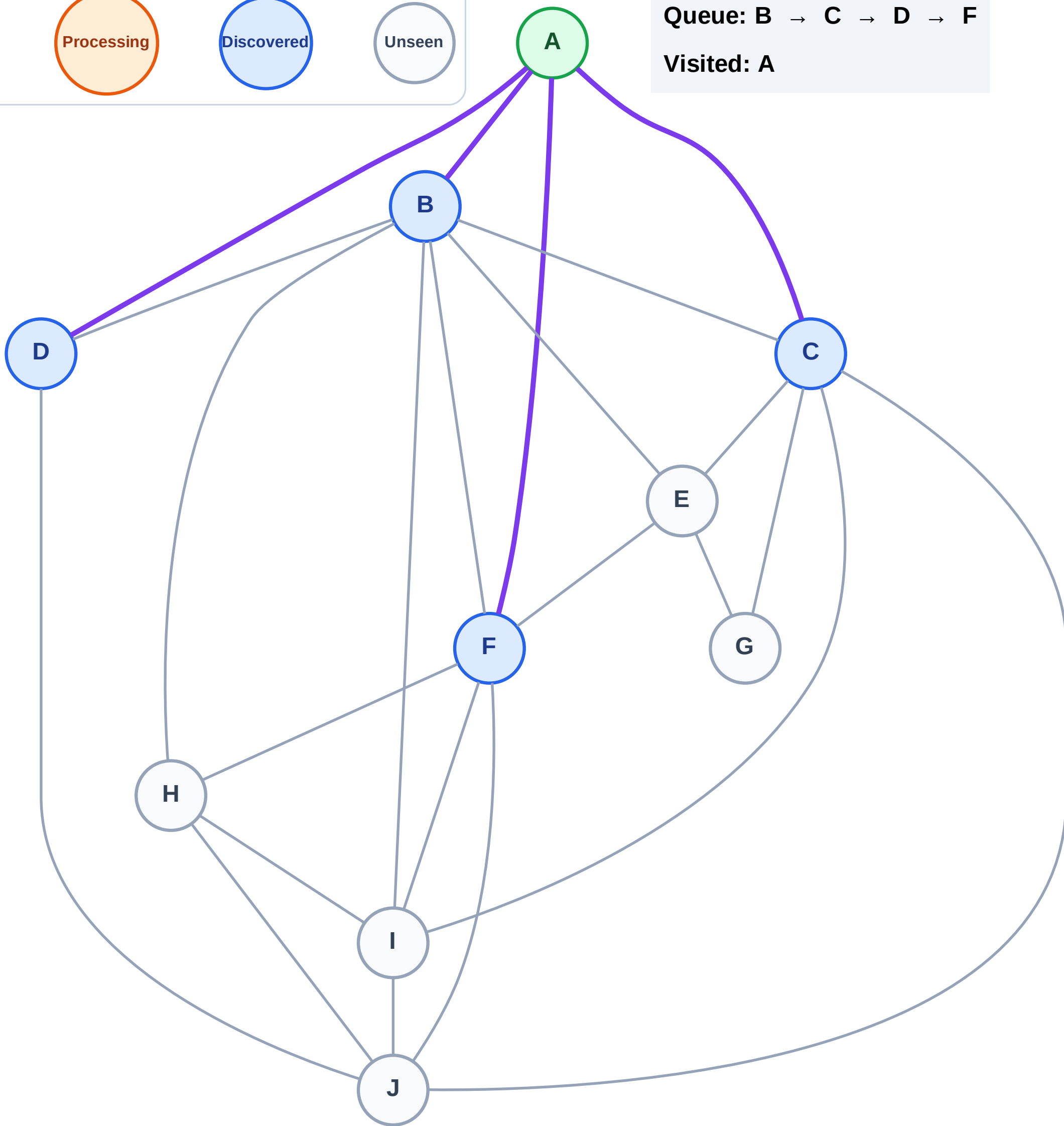
Step 3: Discover B, C, D, F from A

Legend



Queue: B → C → D → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

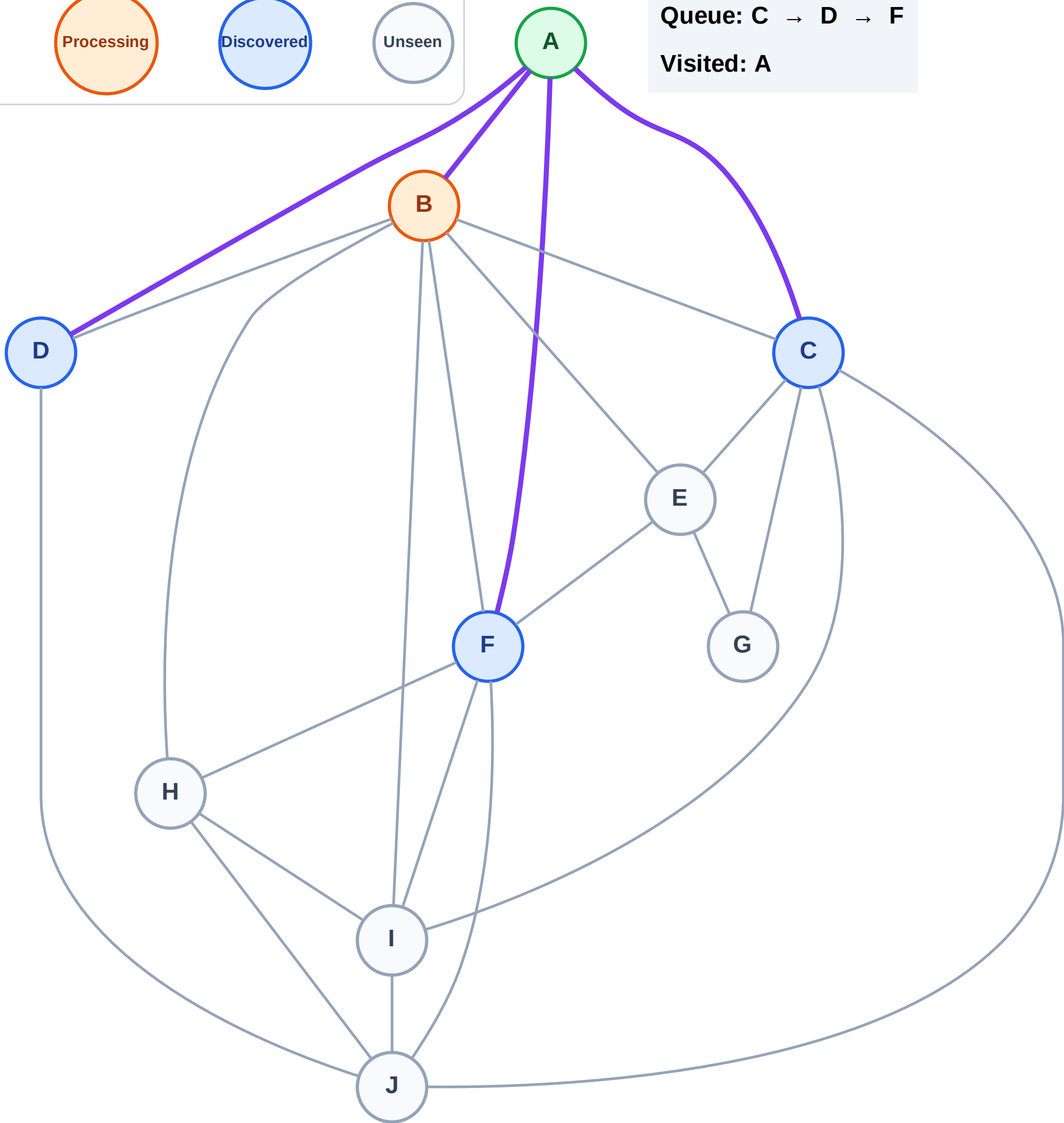
Processing

Discovered

Unseen

Queue: C → D → F

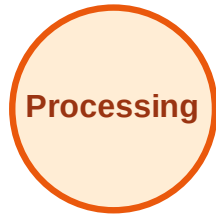
Visited: A



BFS Traversal

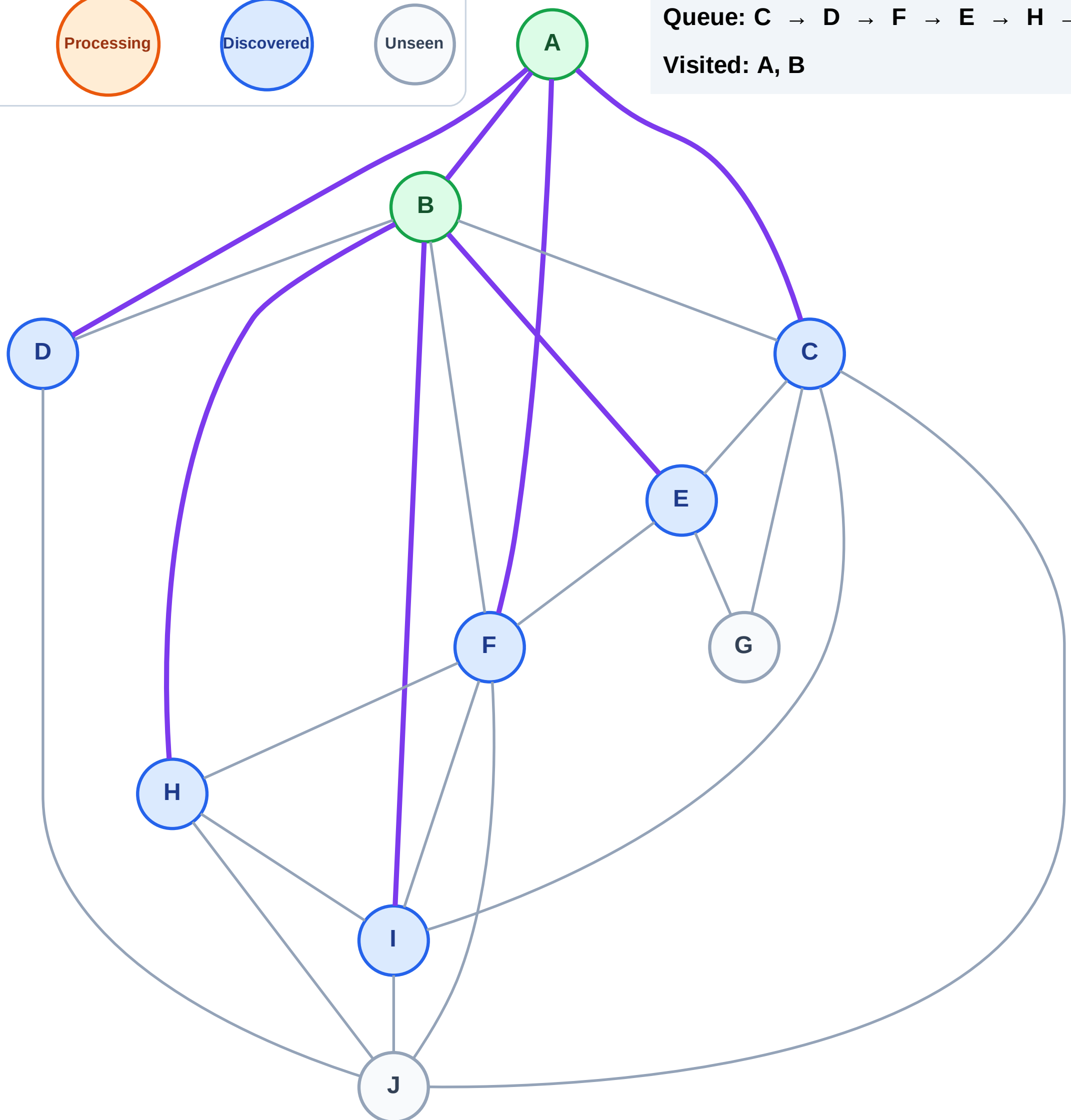
Step 5: Discover E, H, I from B

Legend



Queue: C → D → F → E → H → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

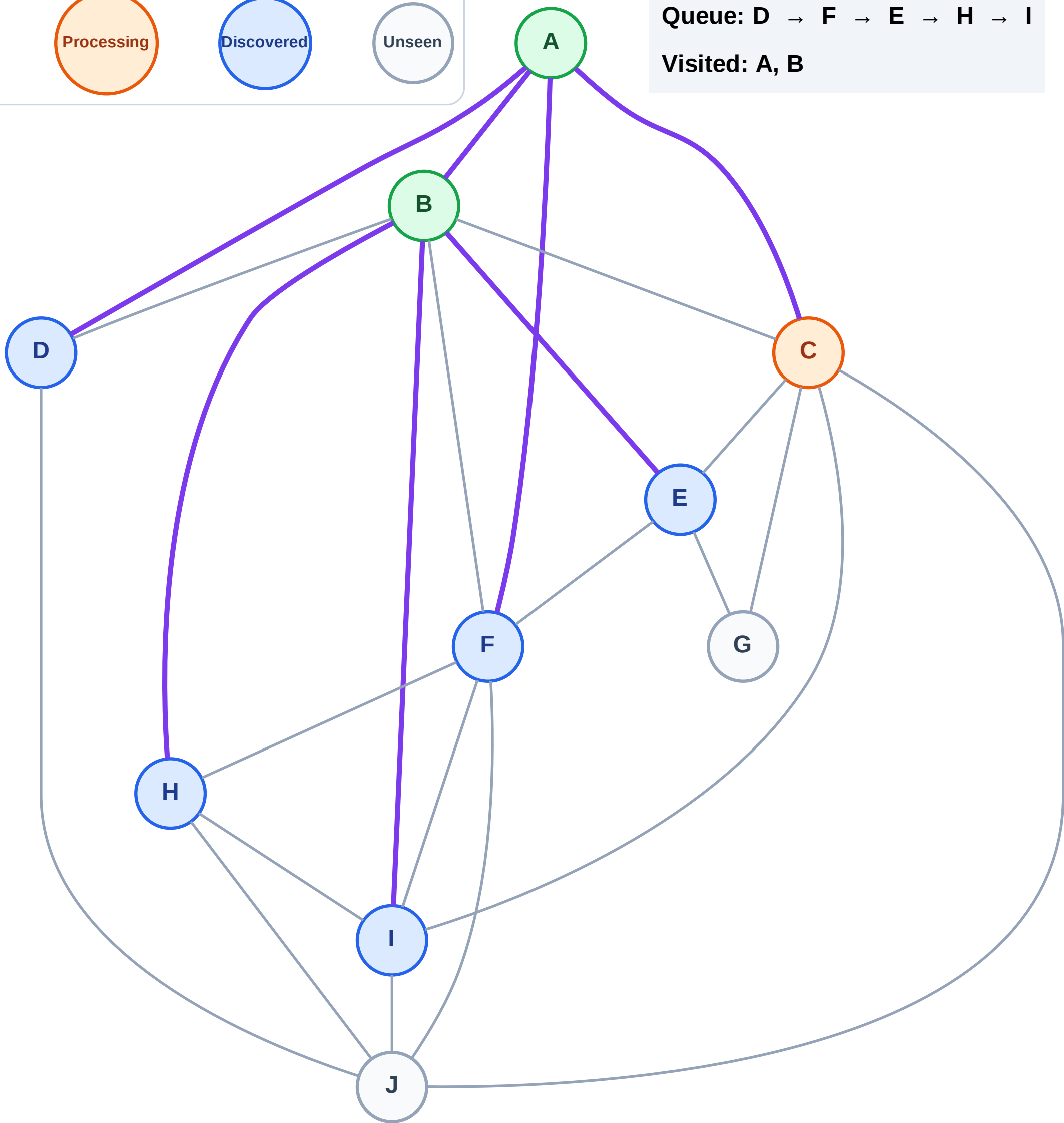
Processing

Discovered

Unseen

Queue: D → F → E → H → I

Visited: A, B



BFS Traversal

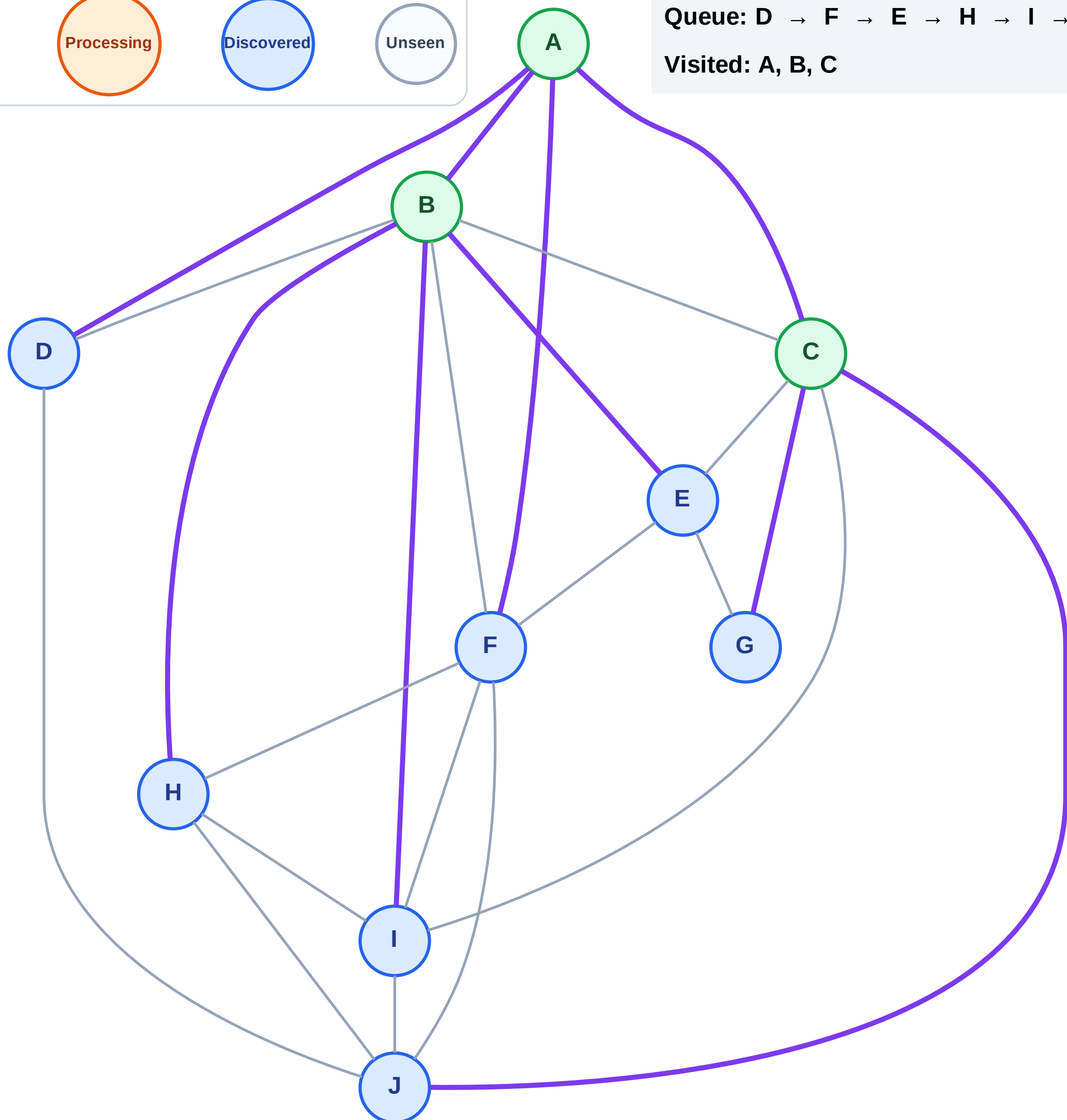
Step 7: Discover G, J from C

Legend



Queue: D → F → E → H → I → G → J

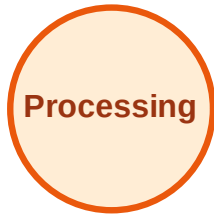
Visited: A, B, C



BFS Traversal

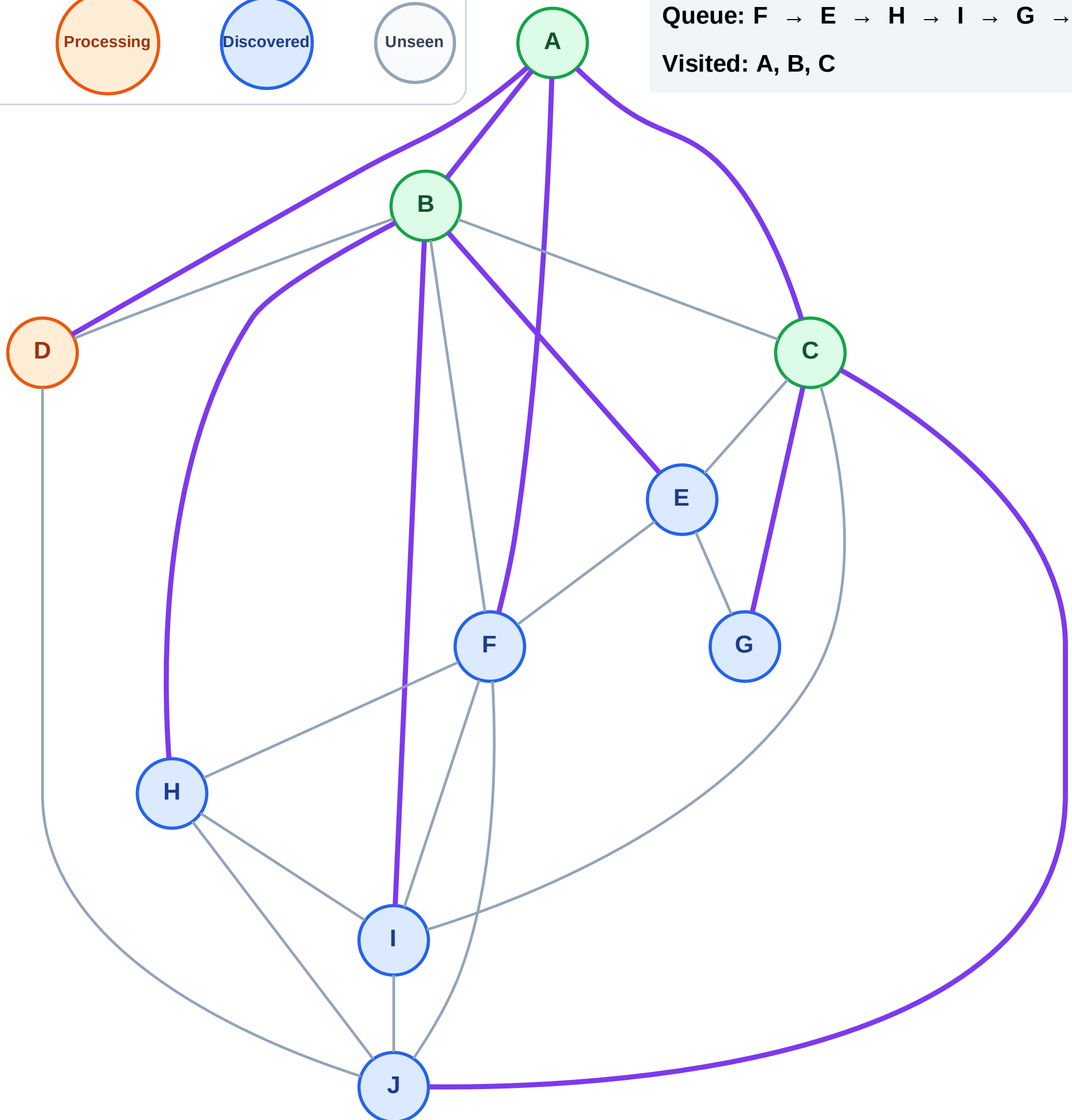
Step 8: Process node D

Legend



Queue: F → E → H → I → G → J

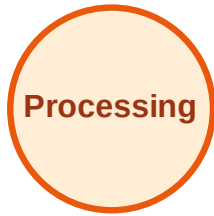
Visited: A, B, C



BFS Traversal

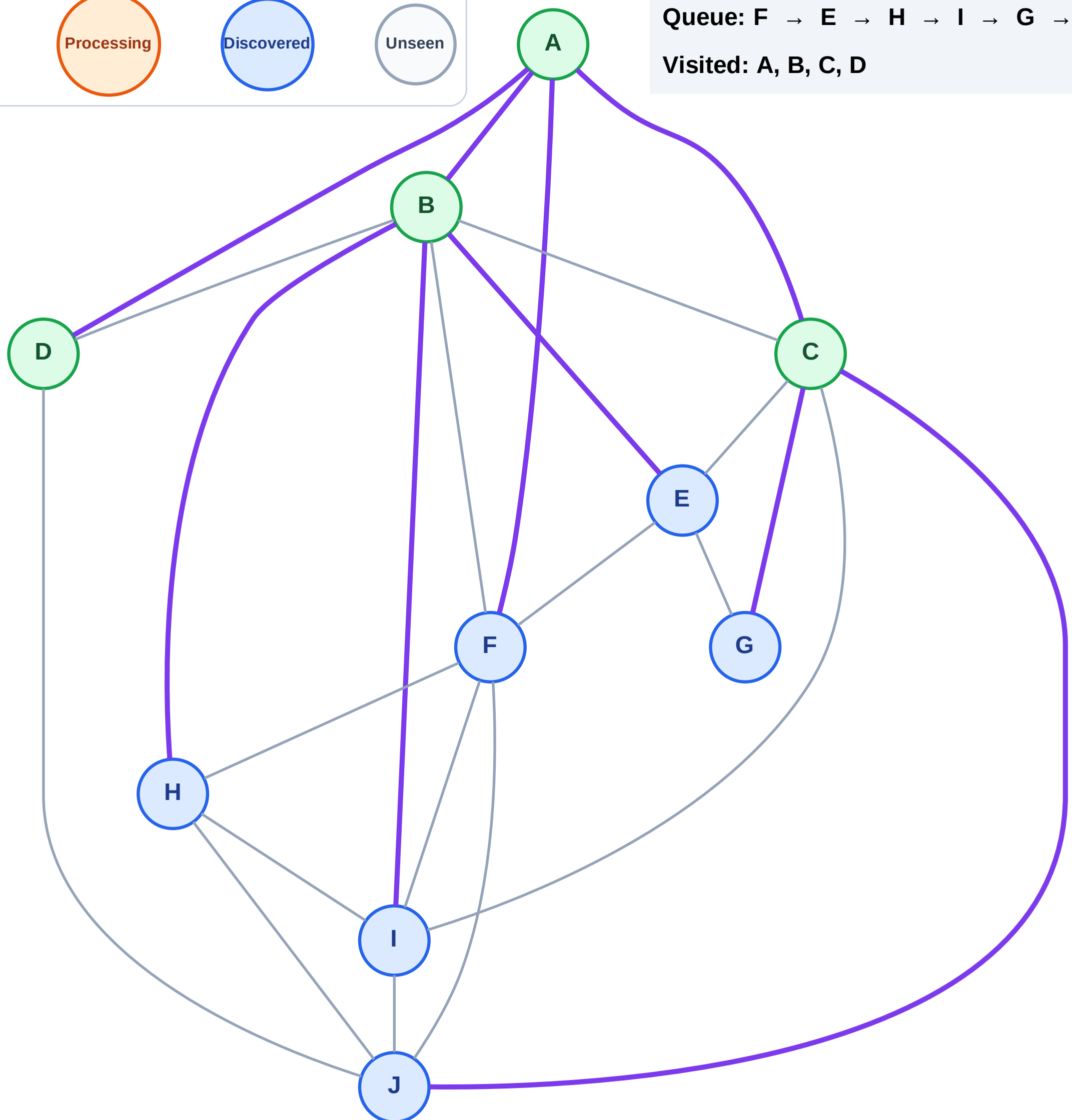
Step 9: No new neighbors from D

Legend



Queue: F → E → H → I → G → J

Visited: A, B, C, D



BFS Traversal

Step 10: Process node F

Legend

Complete

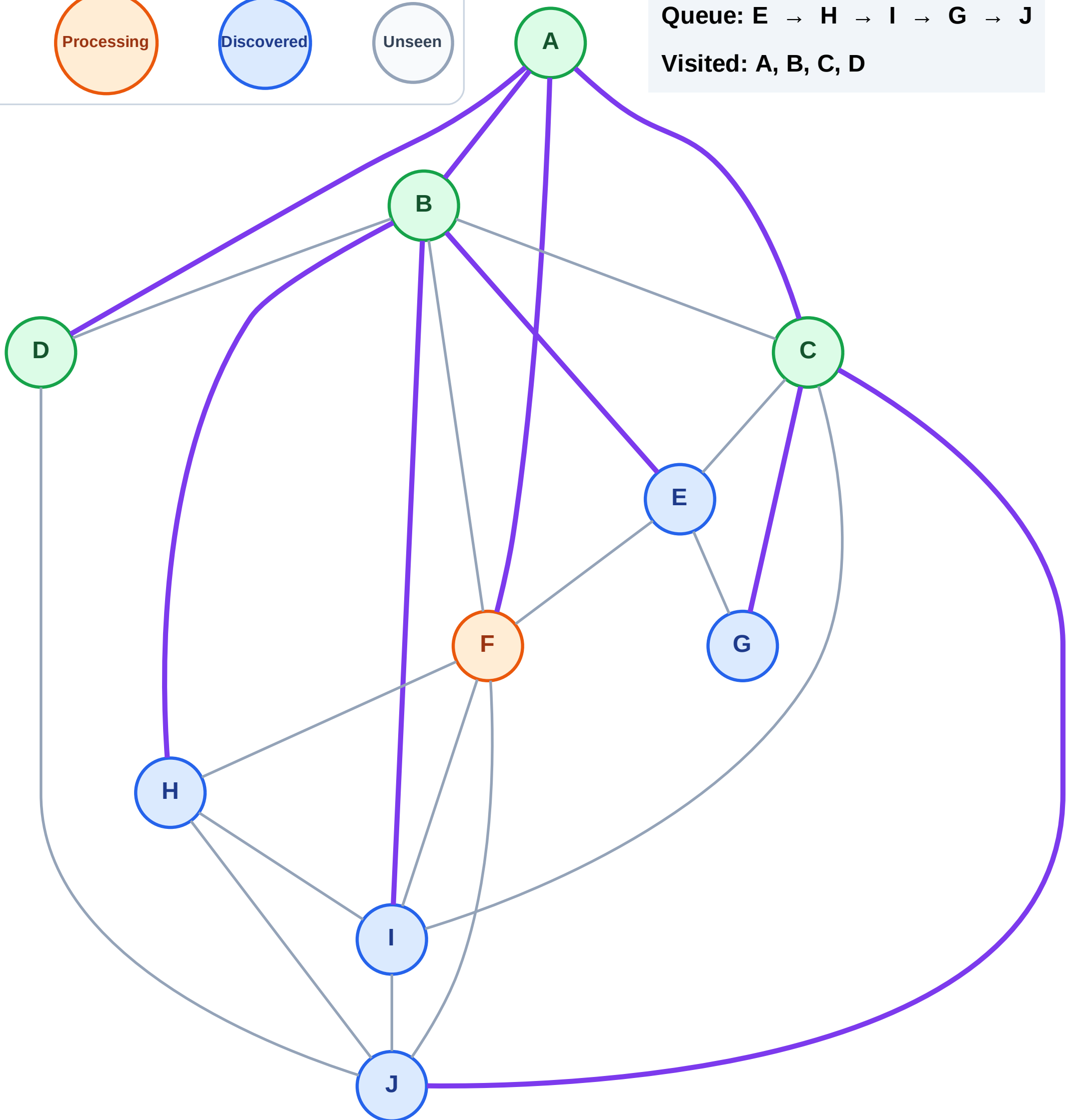
Processing

Discovered

Unseen

Queue: E → H → I → G → J

Visited: A, B, C, D



BFS Traversal

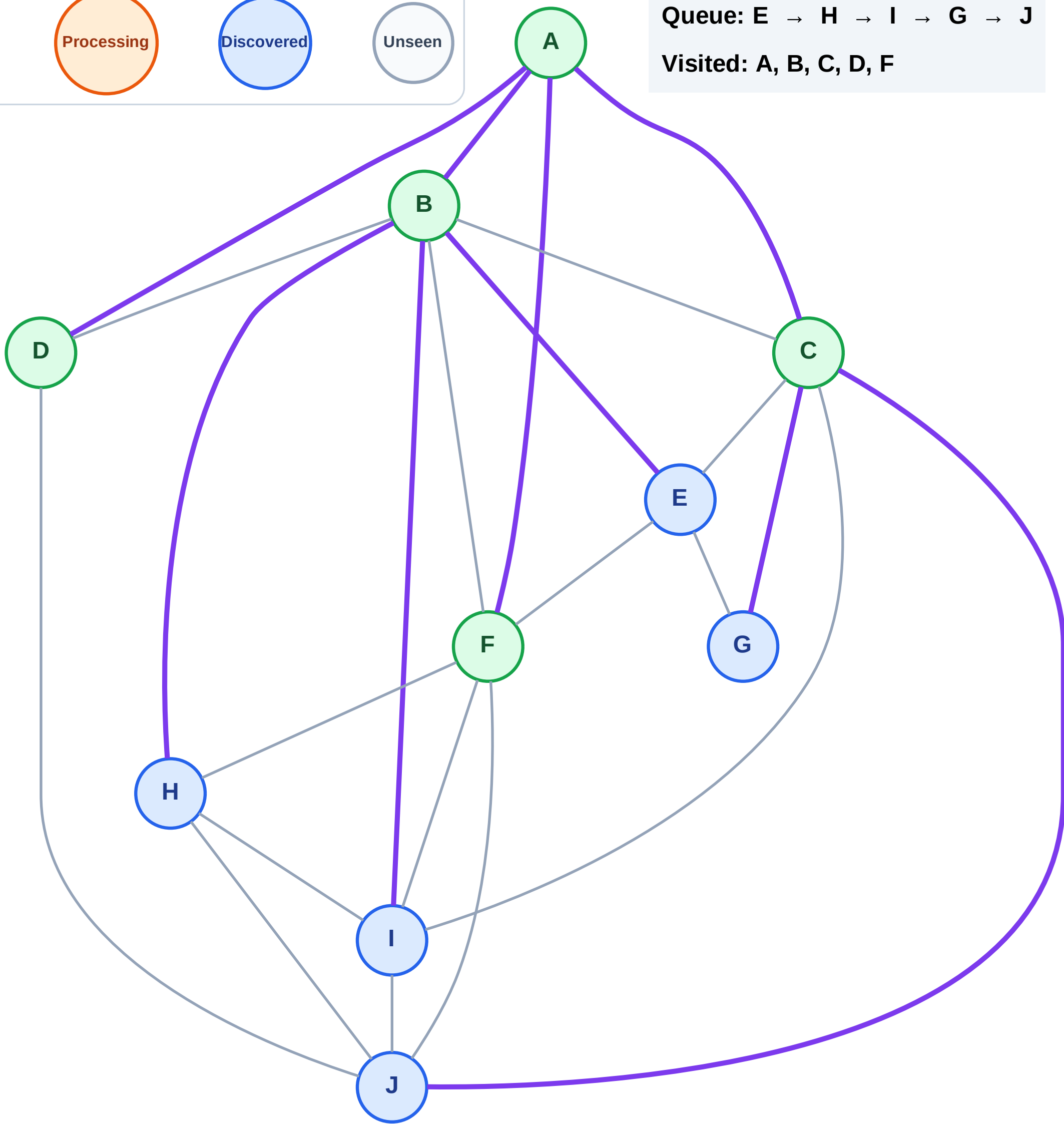
Step 11: No new neighbors from F

Legend



Queue: E → H → I → G → J

Visited: A, B, C, D, F



BFS Traversal

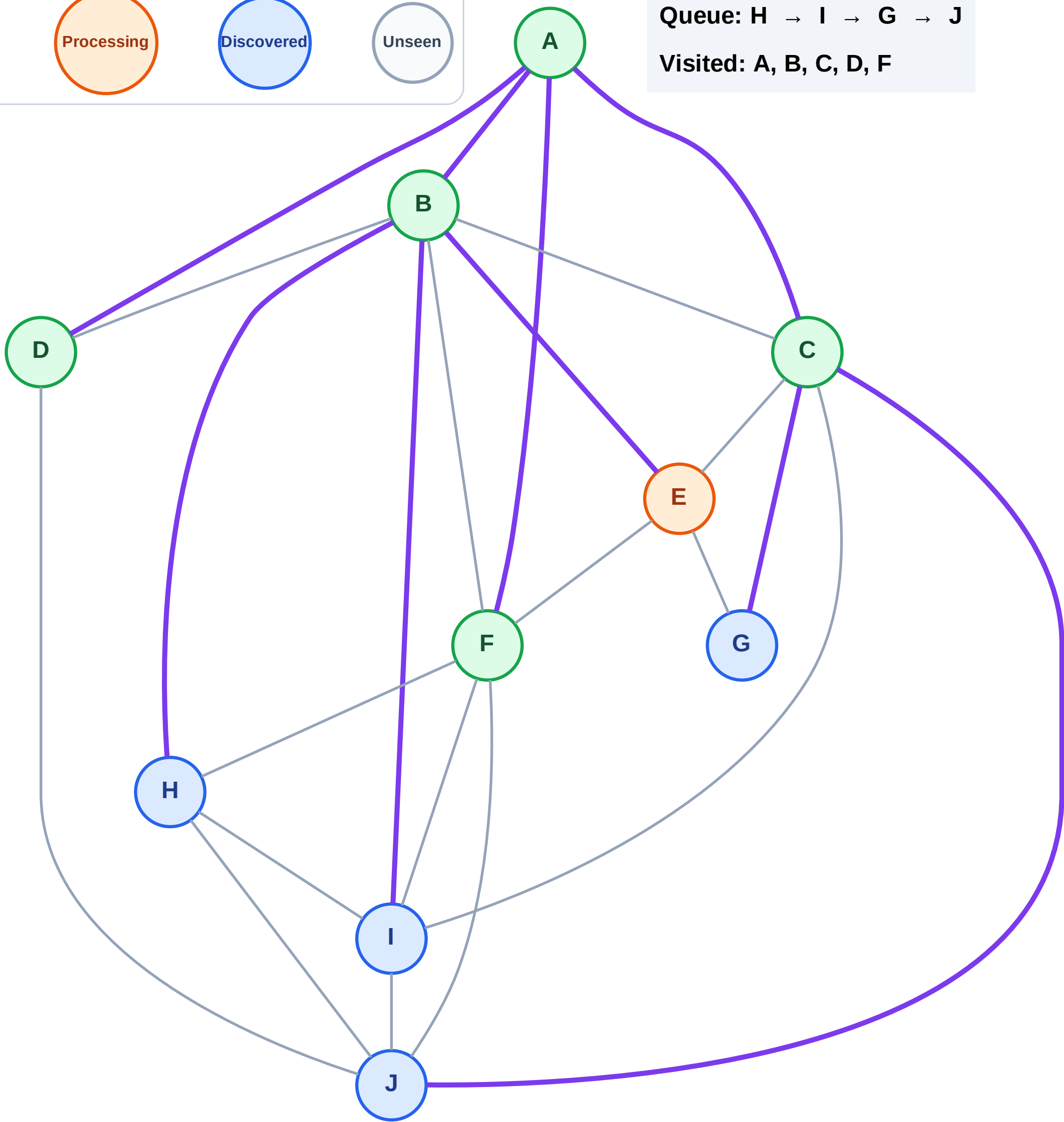
Step 12: Process node E

Legend



Queue: H → I → G → J

Visited: A, B, C, D, F



BFS Traversal

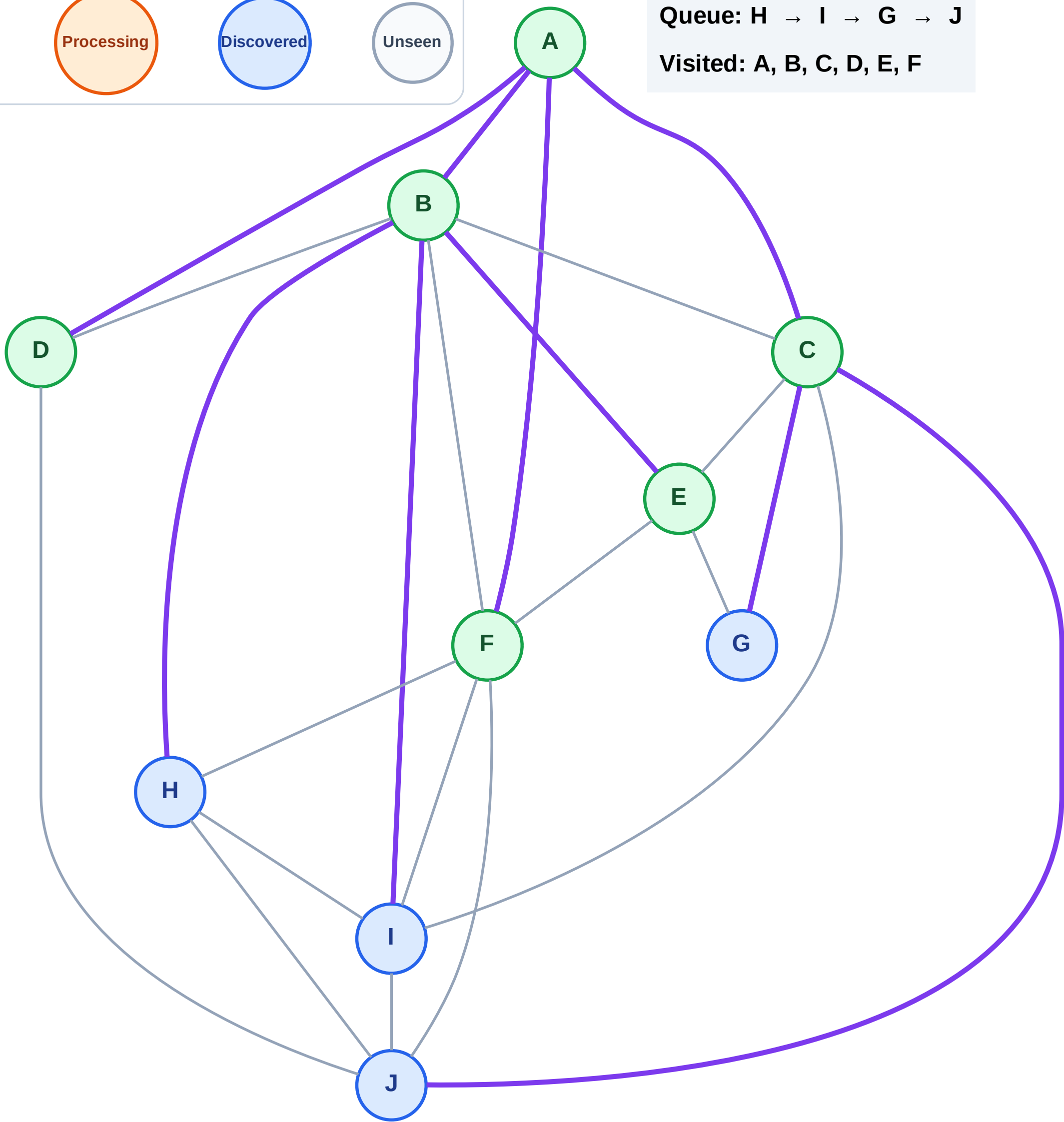
Step 13: No new neighbors from E

Legend



Queue: H → I → G → J

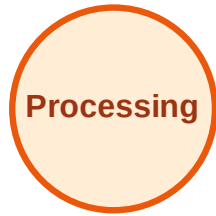
Visited: A, B, C, D, E, F



BFS Traversal

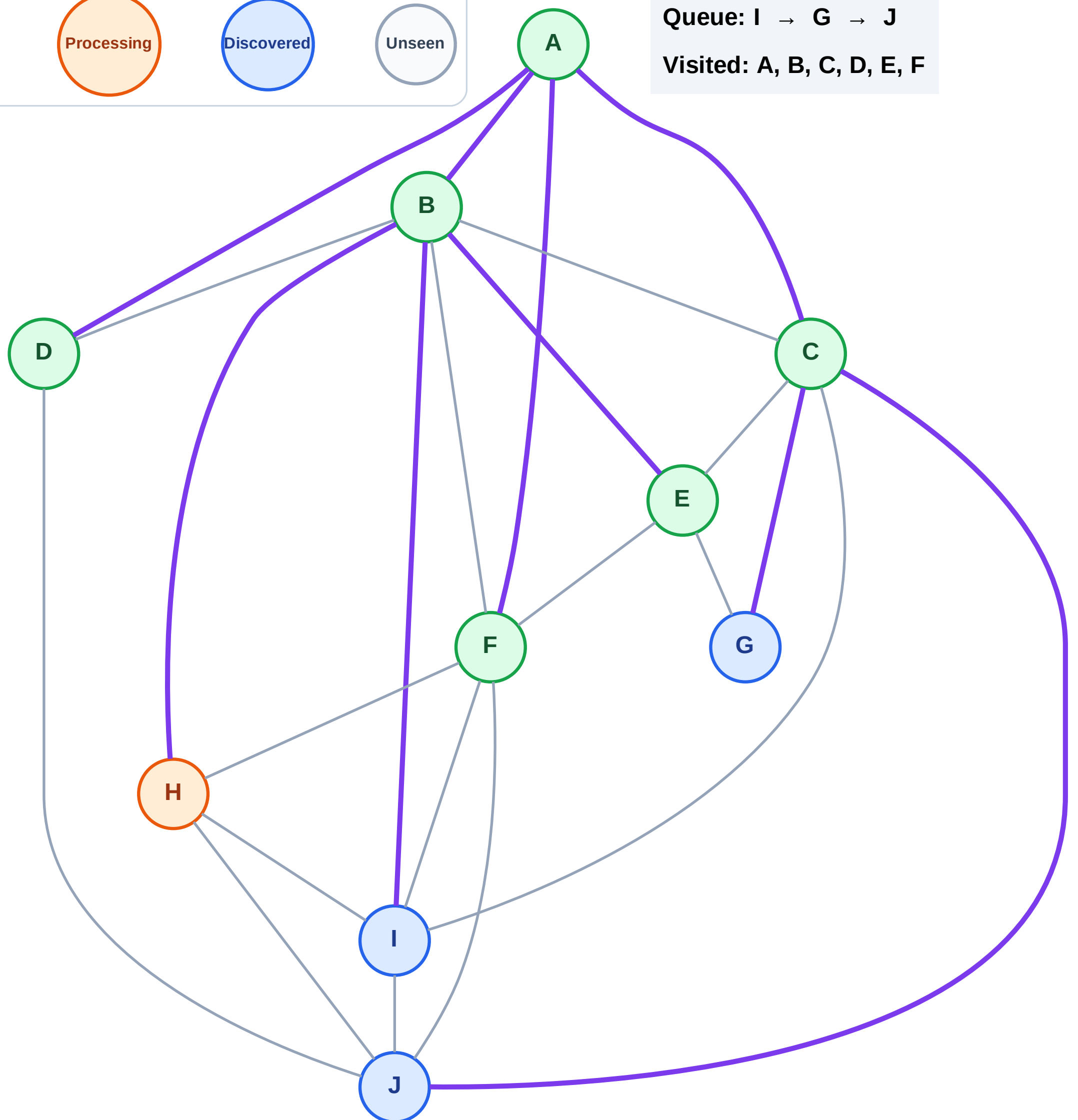
Step 14: Process node H

Legend



Queue: I → G → J

Visited: A, B, C, D, E, F



BFS Traversal

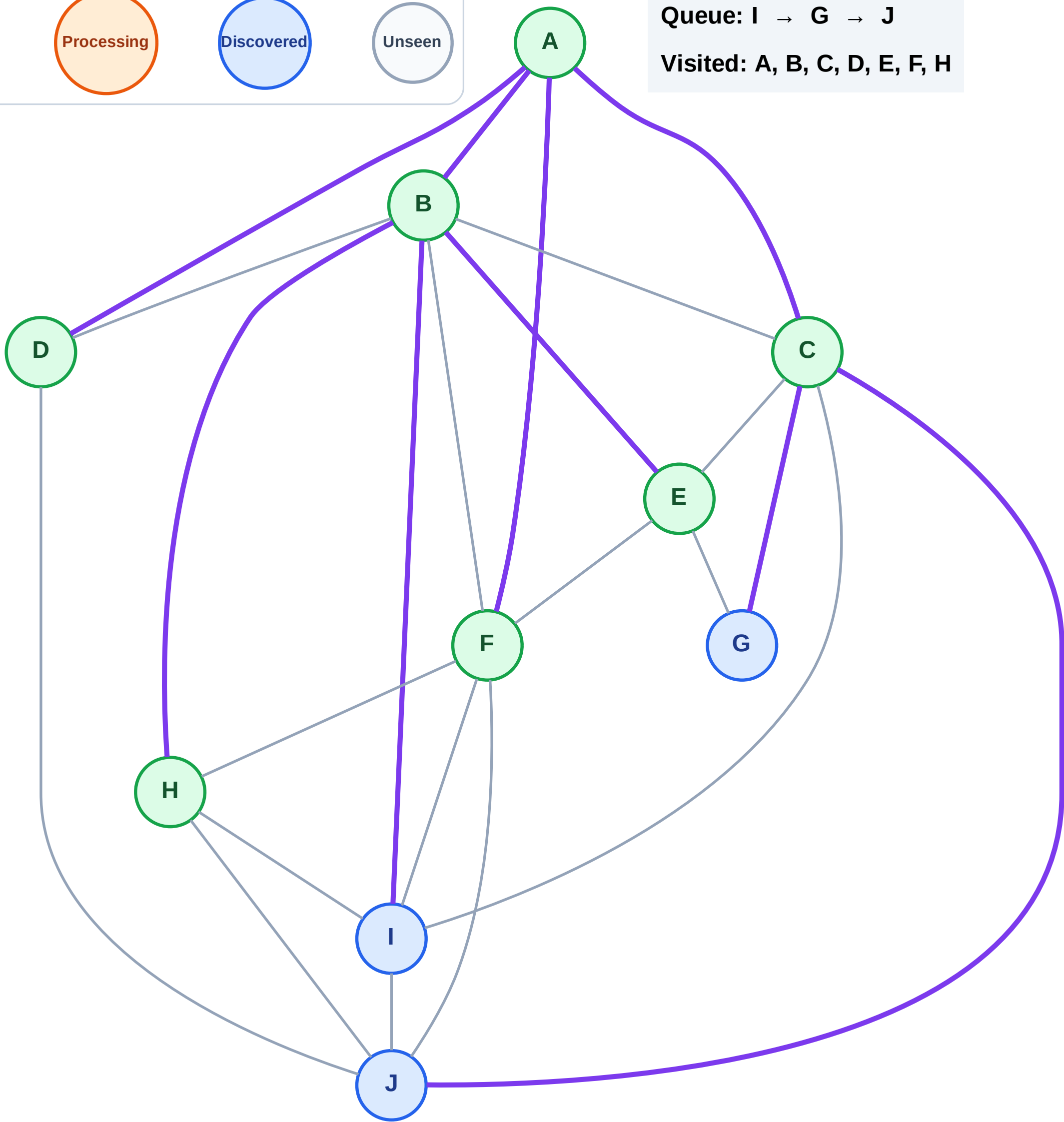
Step 15: No new neighbors from H

Legend



Queue: I → G → J

Visited: A, B, C, D, E, F, H



BFS Traversal

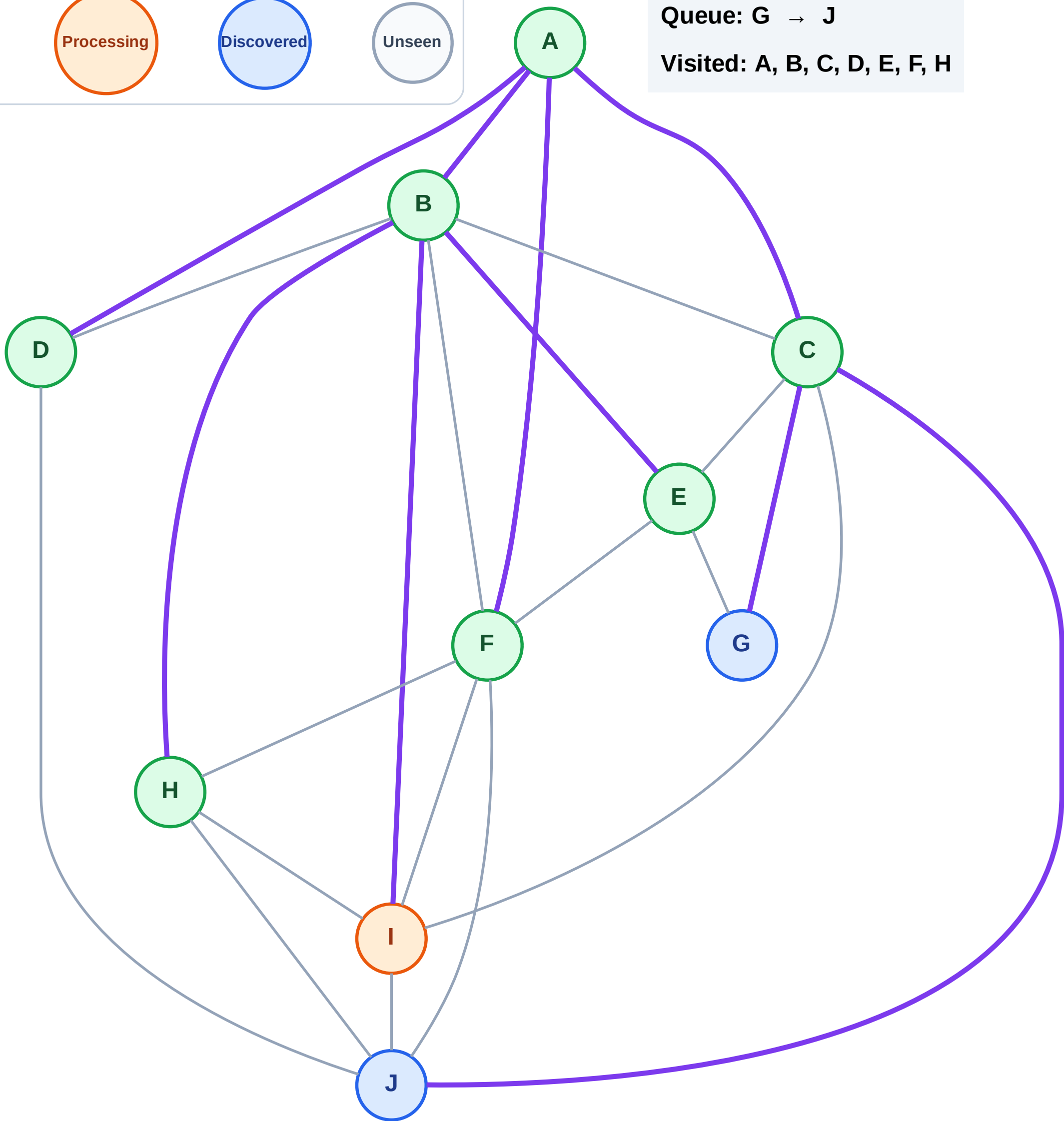
Step 16: Process node I

Legend



Queue: G → J

Visited: A, B, C, D, E, F, H



BFS Traversal

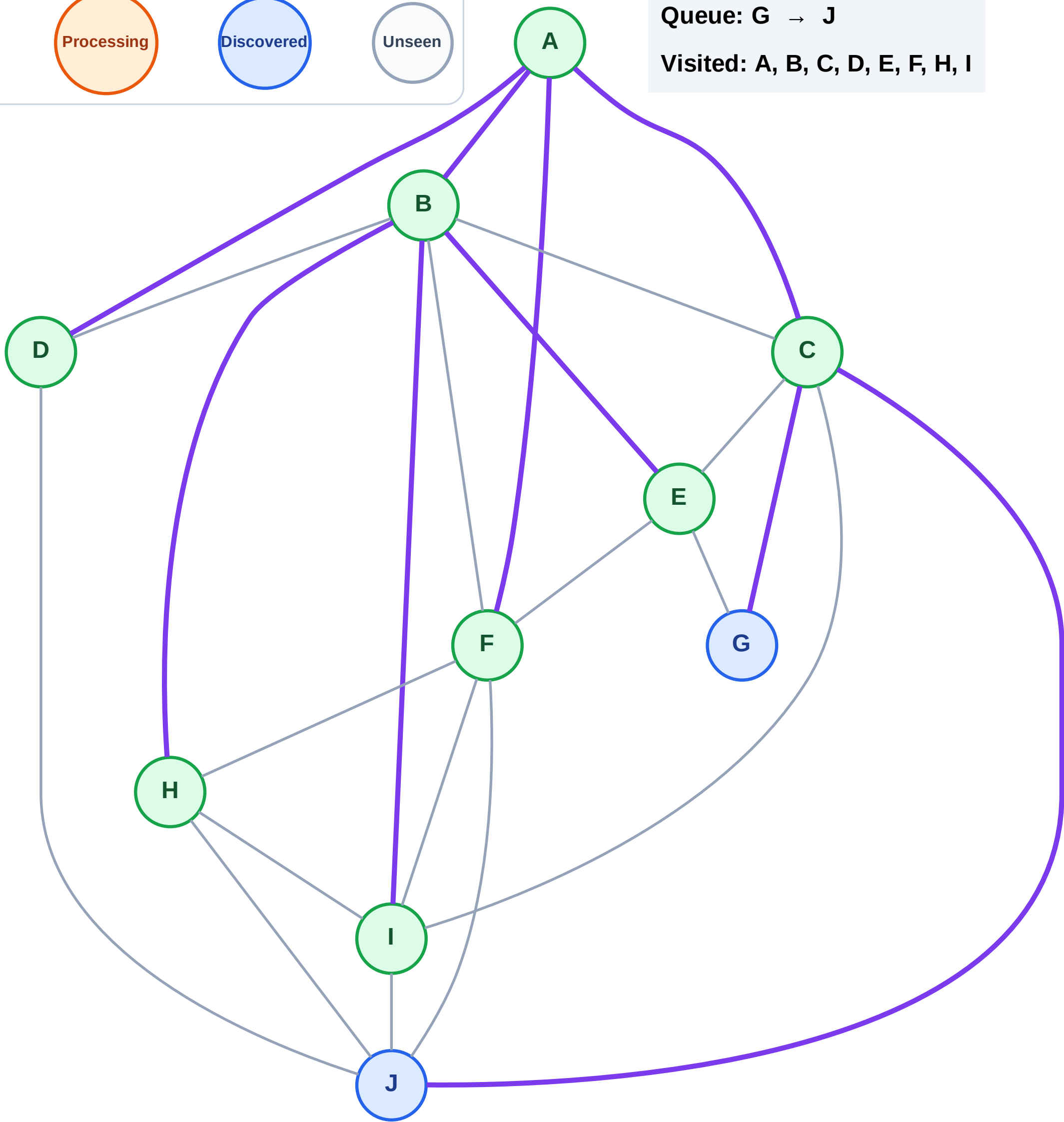
Step 17: No new neighbors from I

Legend



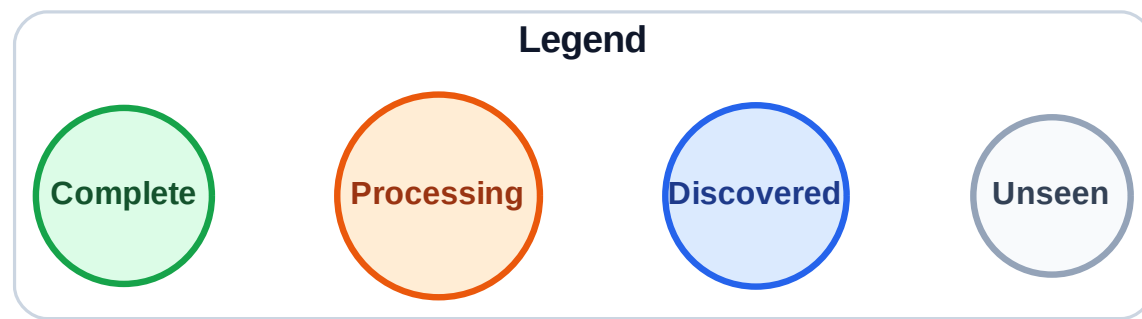
Queue: G → J

Visited: A, B, C, D, E, F, H, I

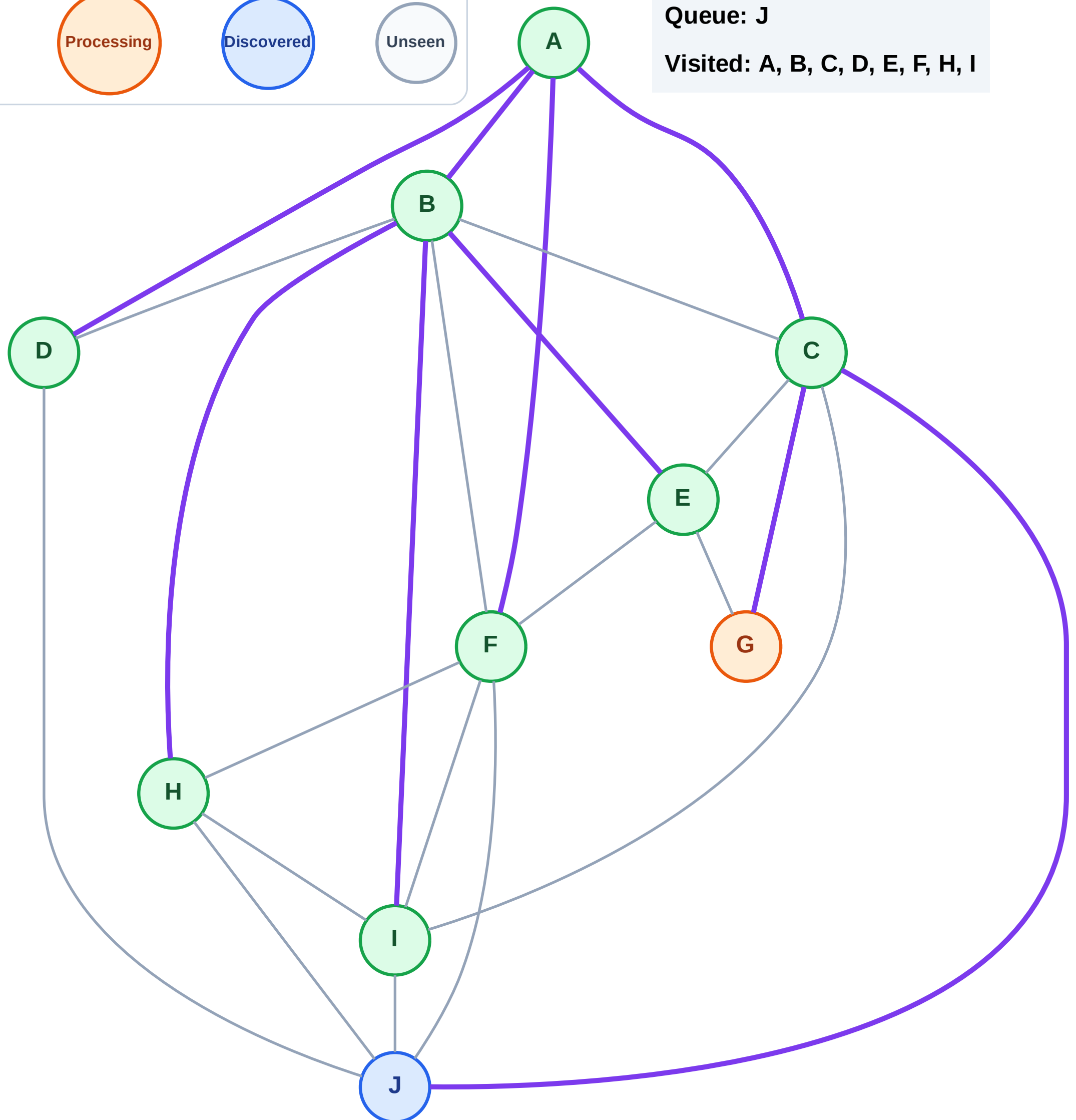


BFS Traversal

Step 18: Process node G



Queue: J
Visited: A, B, C, D, E, F, H, I



BFS Traversal

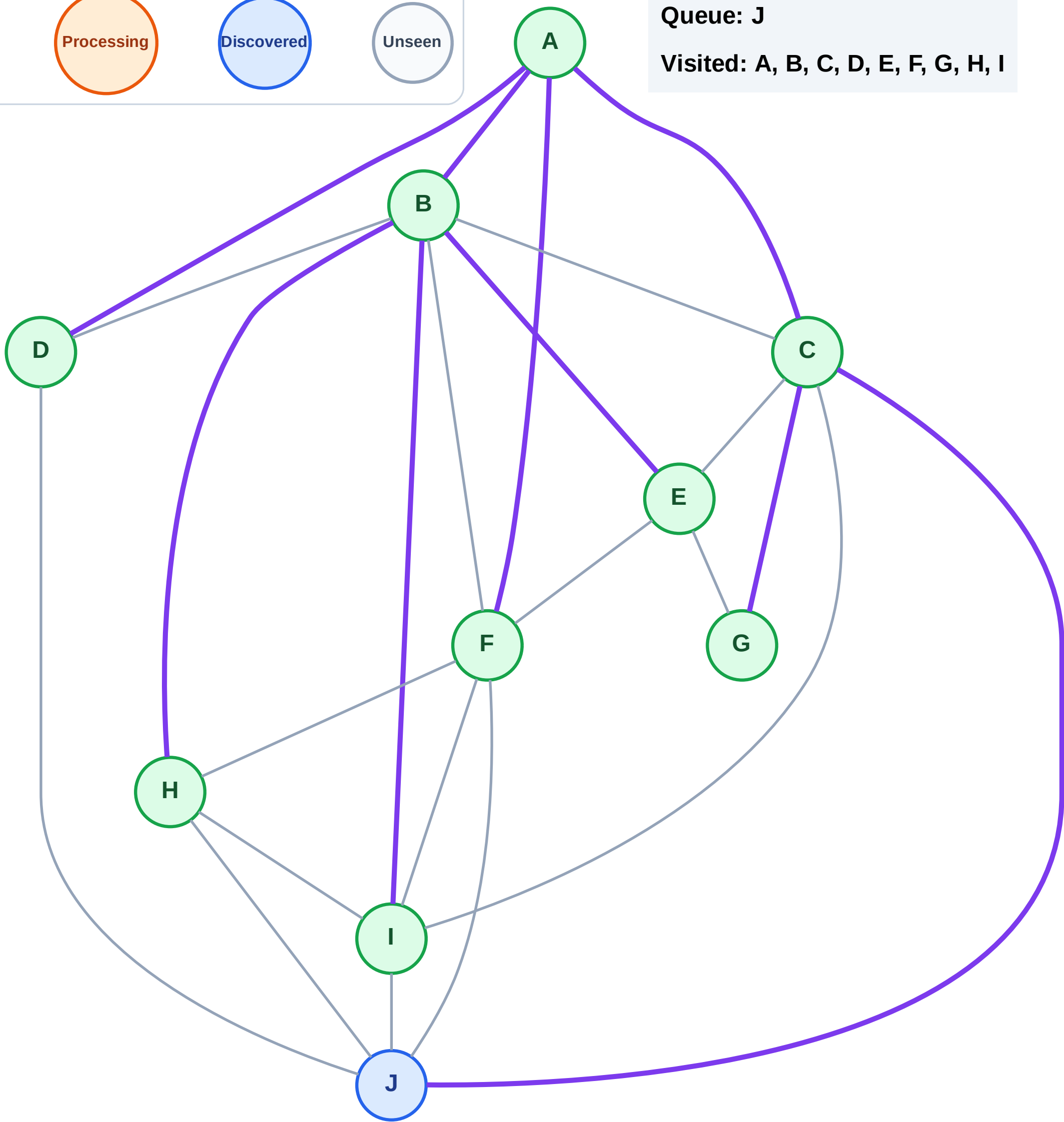
Step 19: No new neighbors from G

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

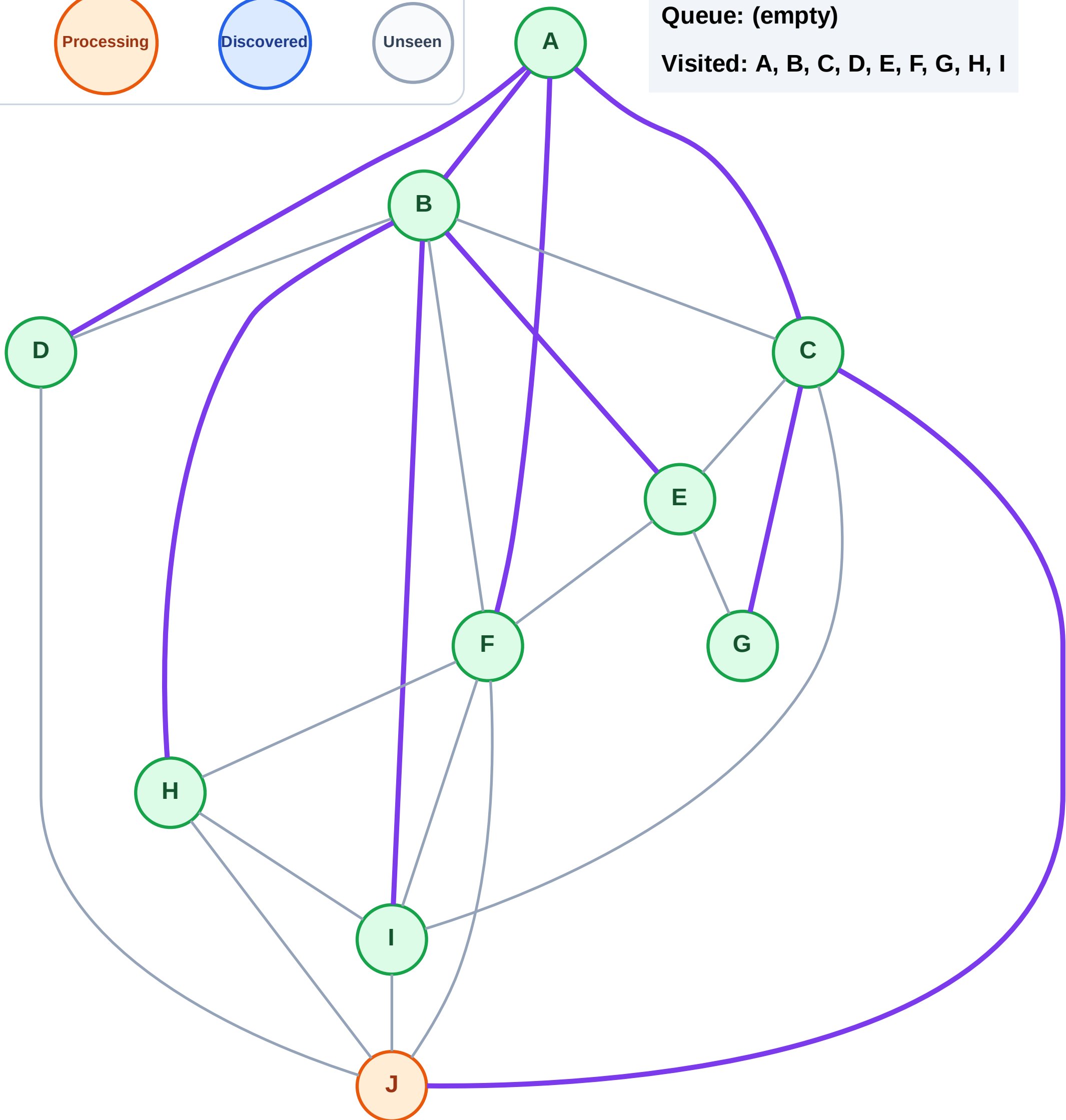
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

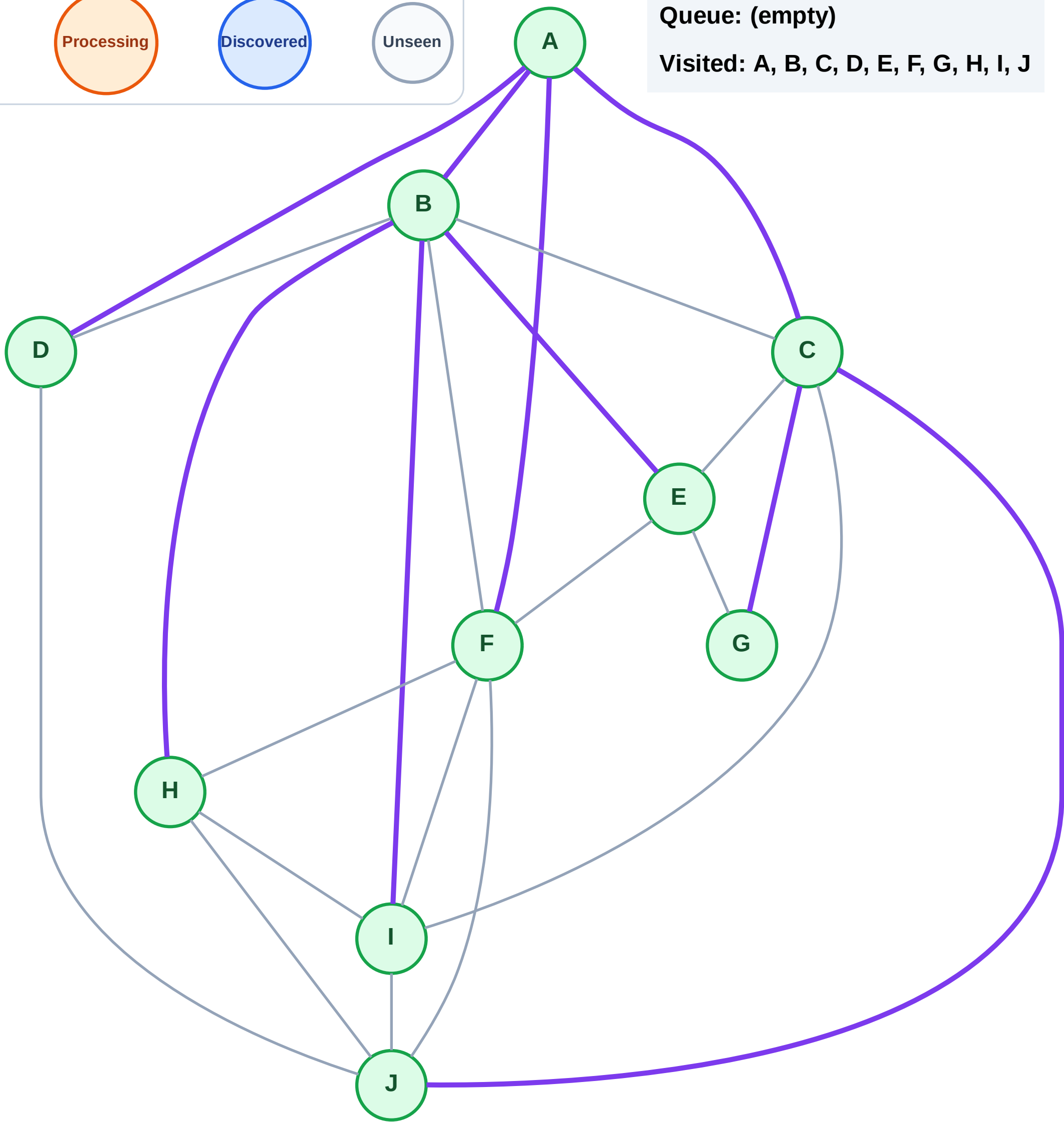
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

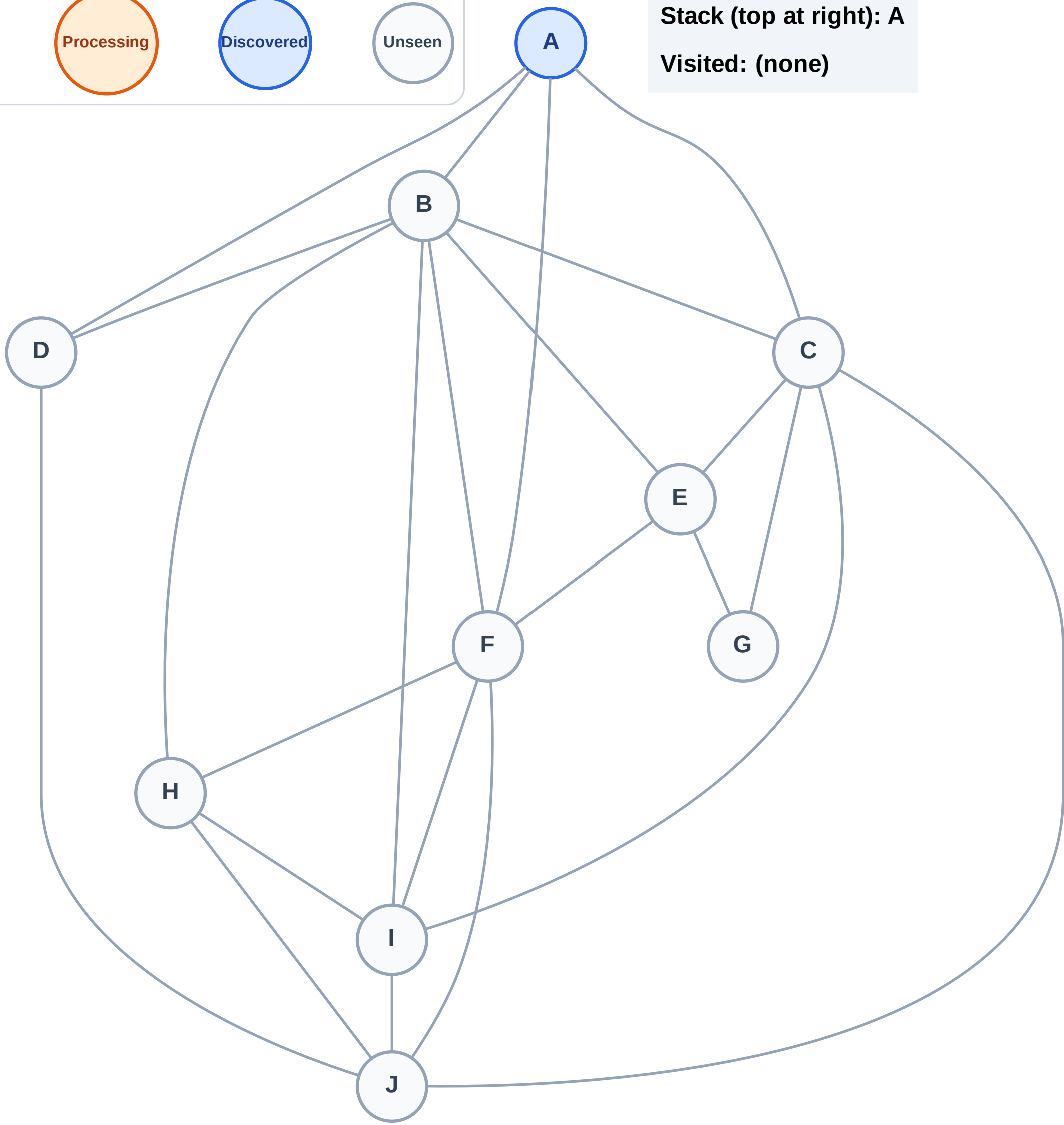
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

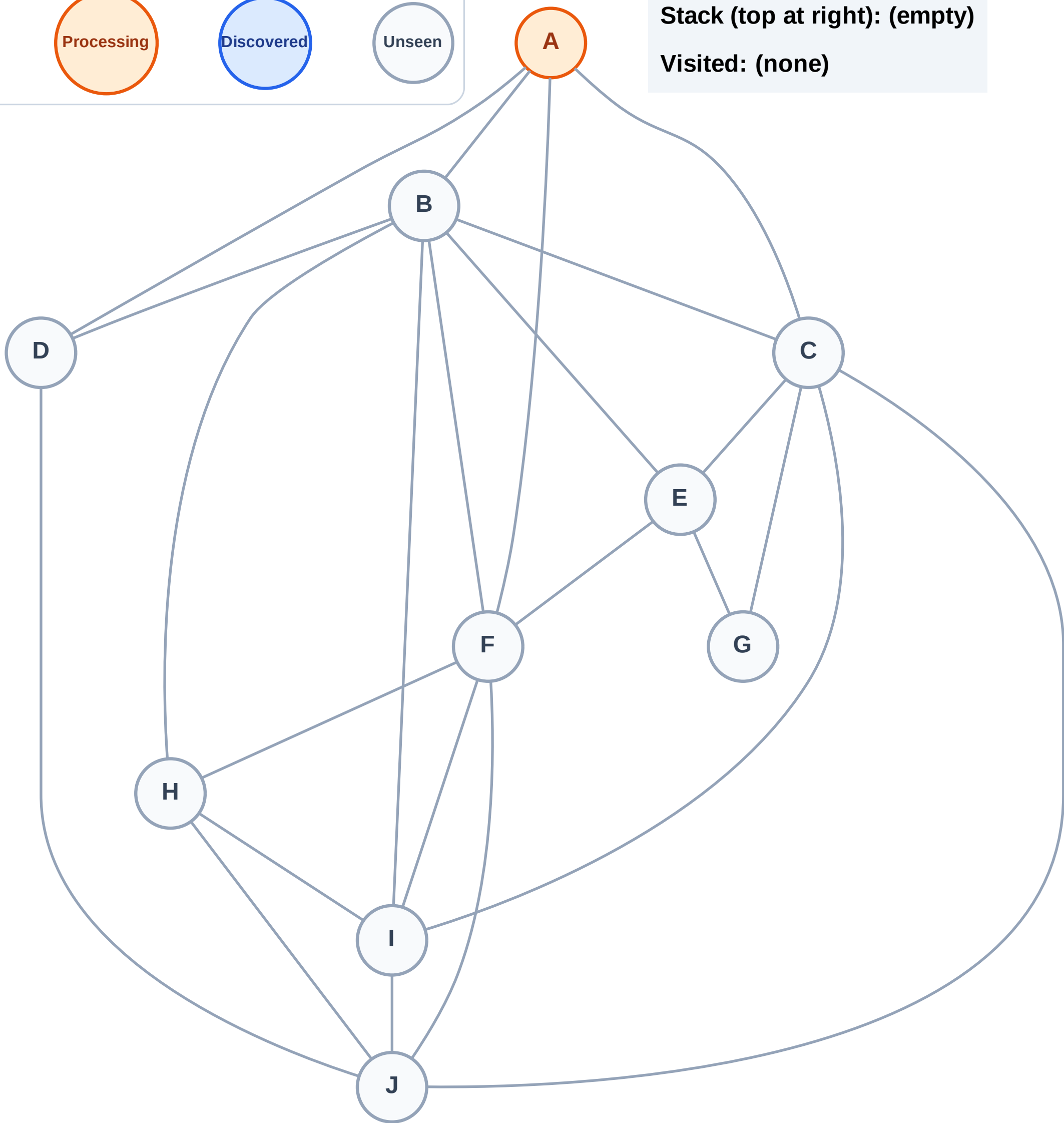
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

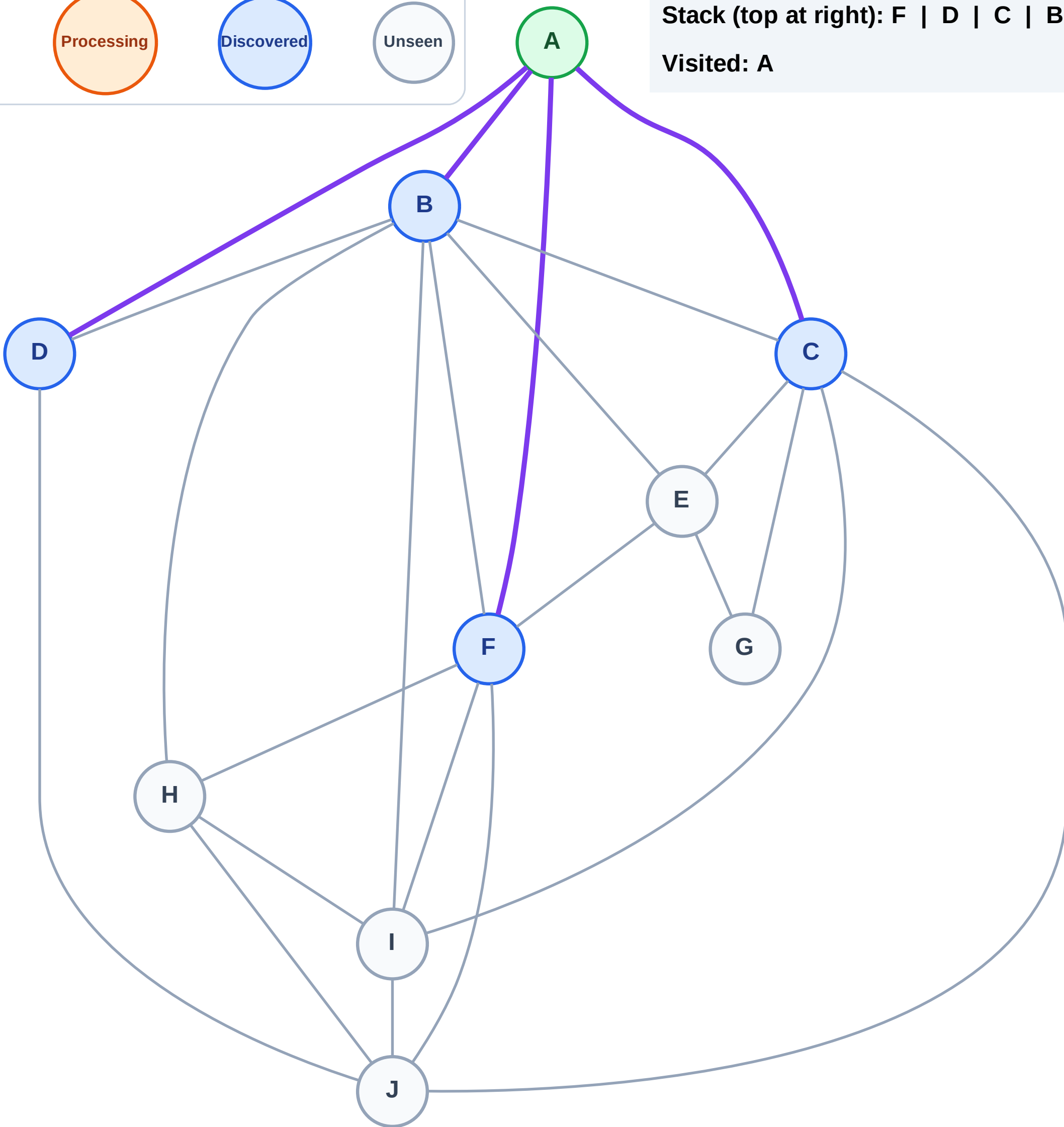
Step 3: Discover F, D, C, B from A

Legend



Stack (top at right): F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

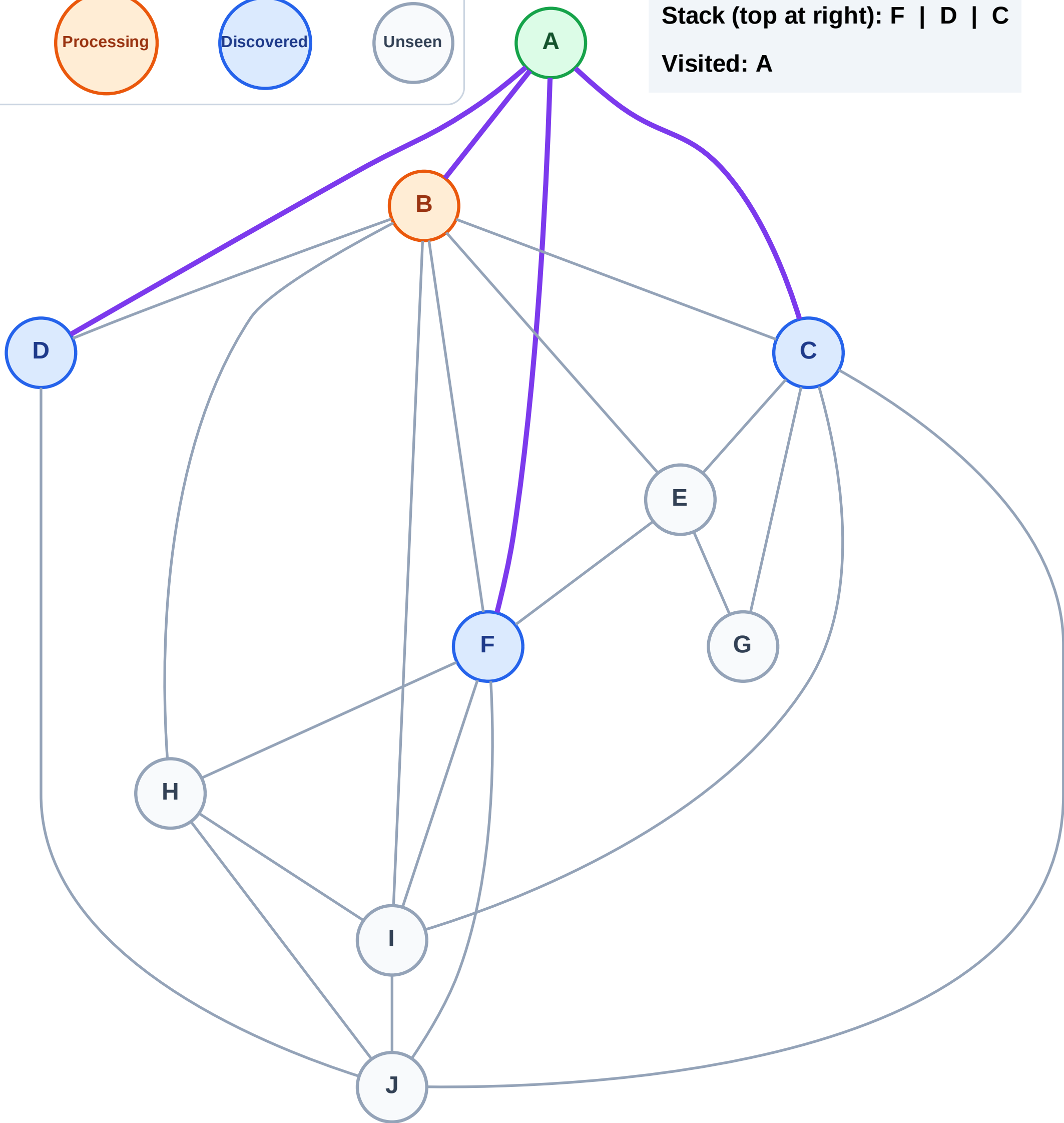
Processing

Discovered

Unseen

Stack (top at right): F | D | C

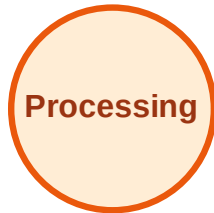
Visited: A



DFS Traversal

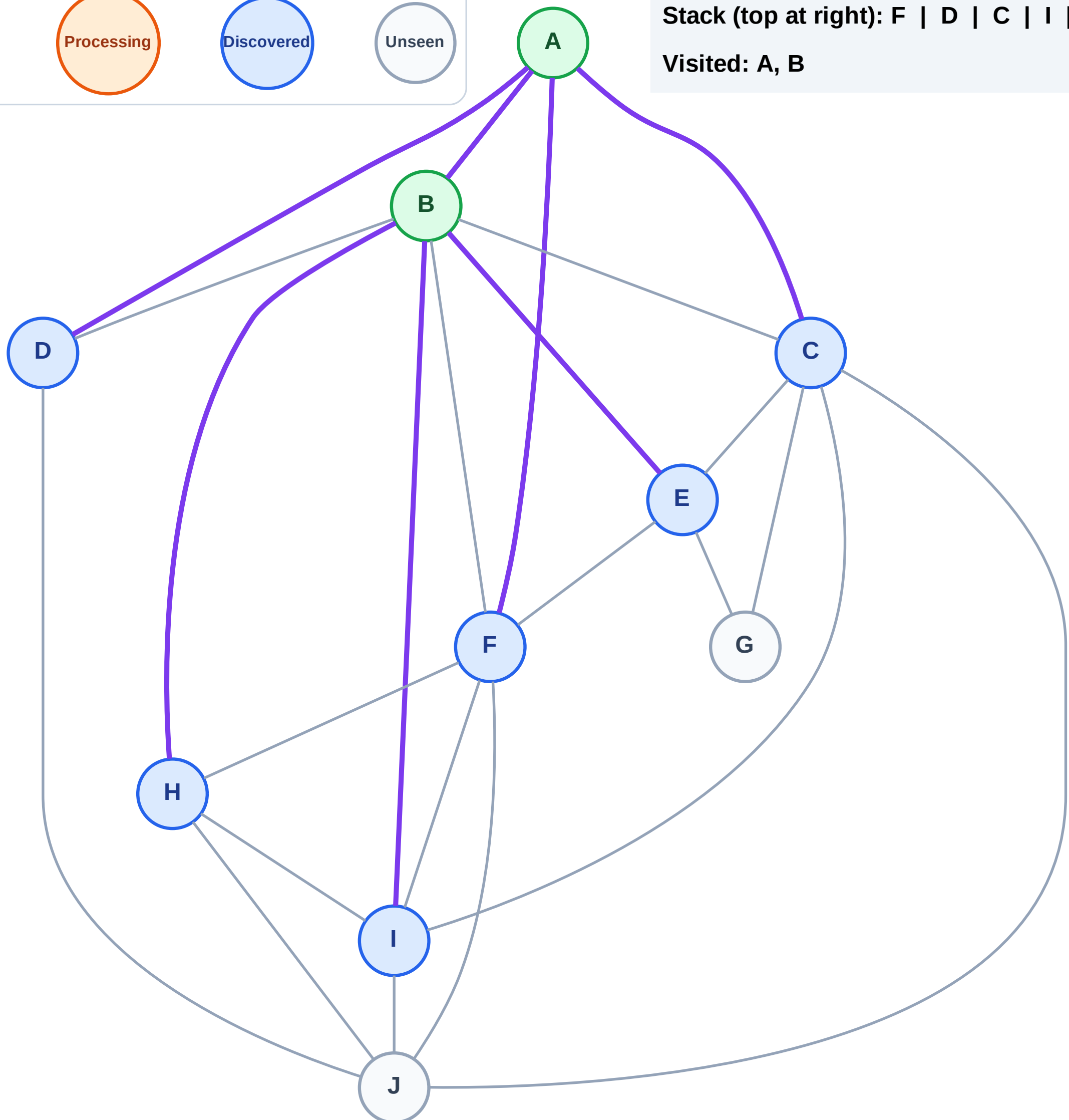
Step 5: Discover I, H, E from B

Legend



Stack (top at right): F | D | C | I | H | E

Visited: A, B



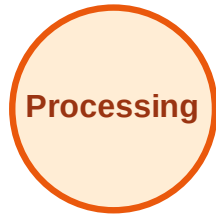
DFS Traversal

Step 6: Process node E

Legend



Complete



Processing



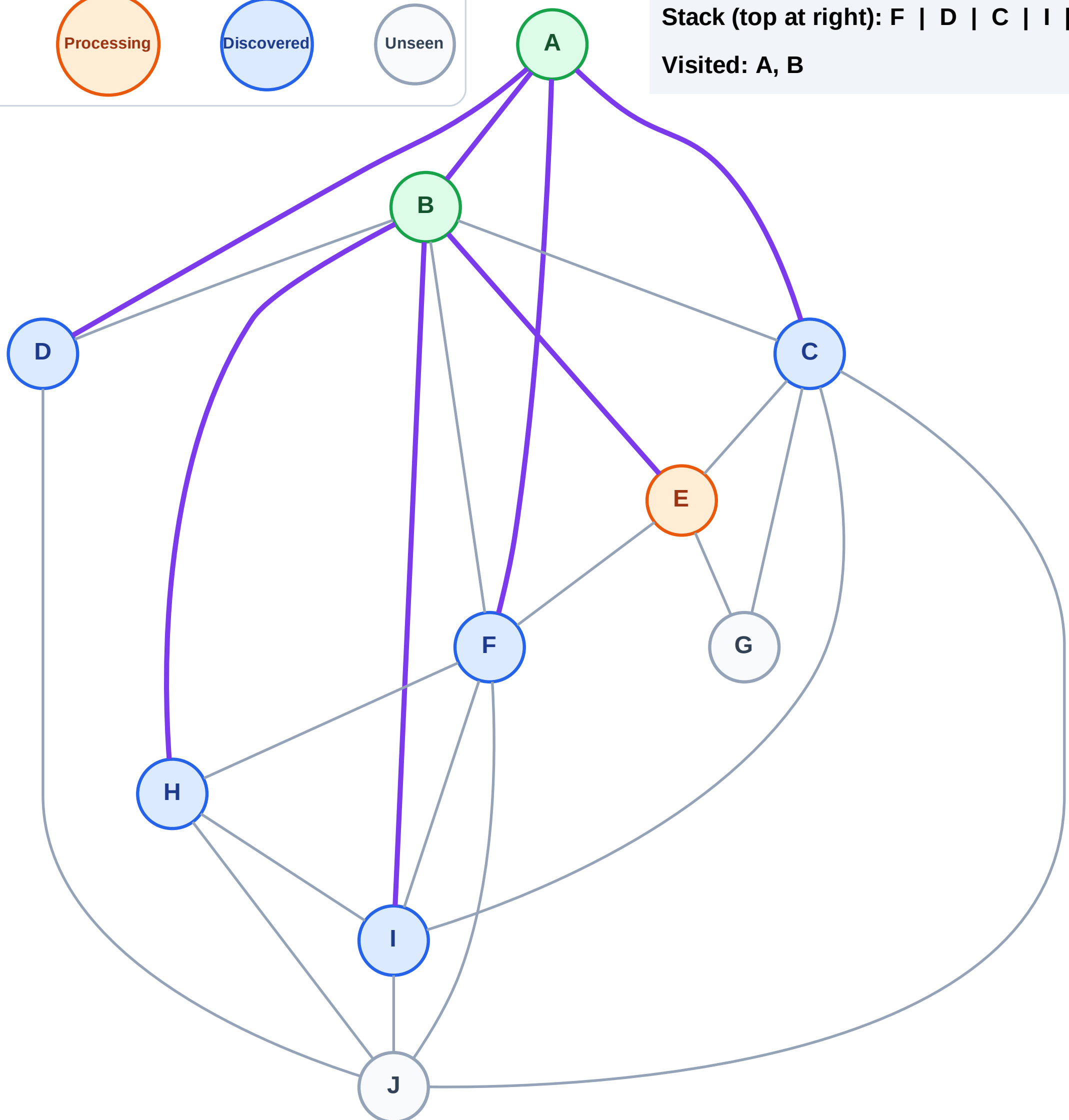
Discovered



Unseen

Stack (top at right): F | D | C | I | H

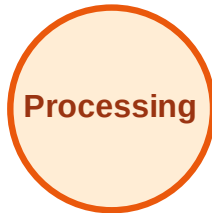
Visited: A, B



DFS Traversal

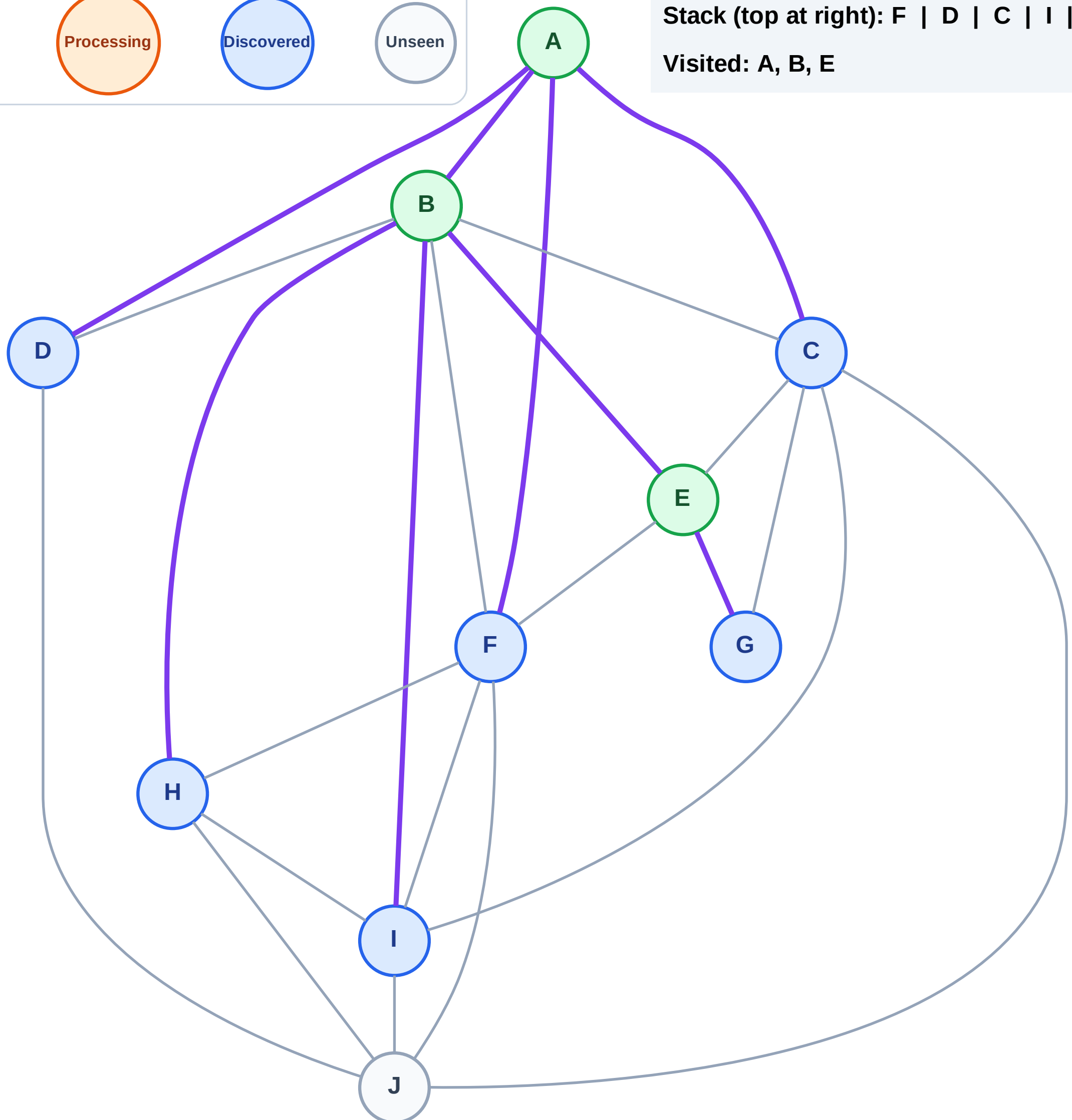
Step 7: Discover G from E

Legend



Stack (top at right): F | D | C | I | H | G

Visited: A, B, E



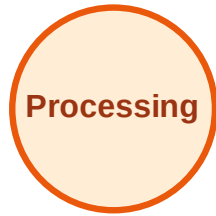
DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



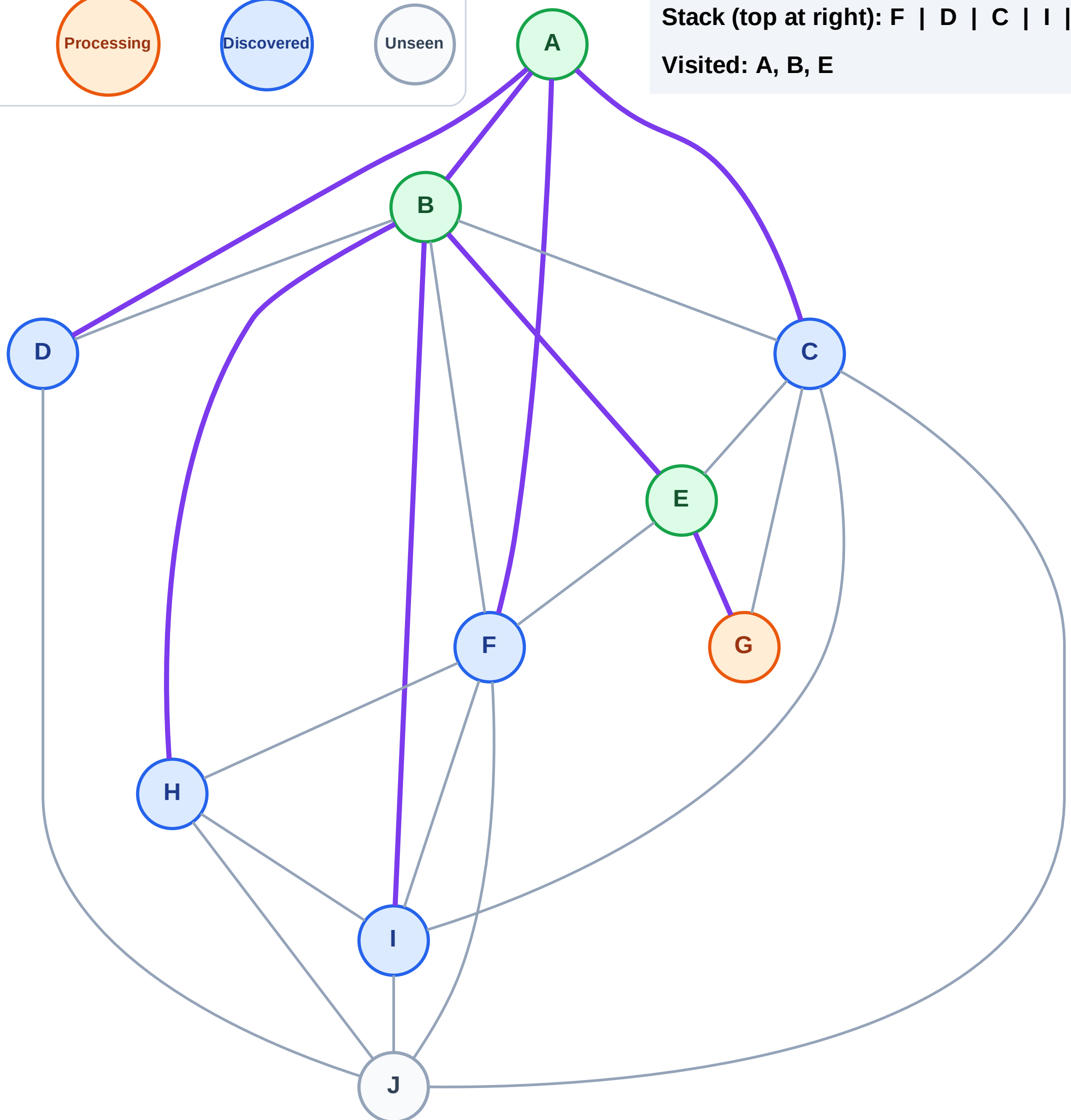
Discovered



Unseen

Stack (top at right): F | D | C | I | H

Visited: A, B, E



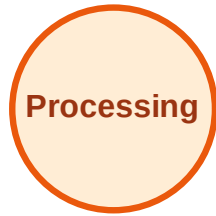
DFS Traversal

Step 9: No new neighbors from G

Legend



Complete



Processing



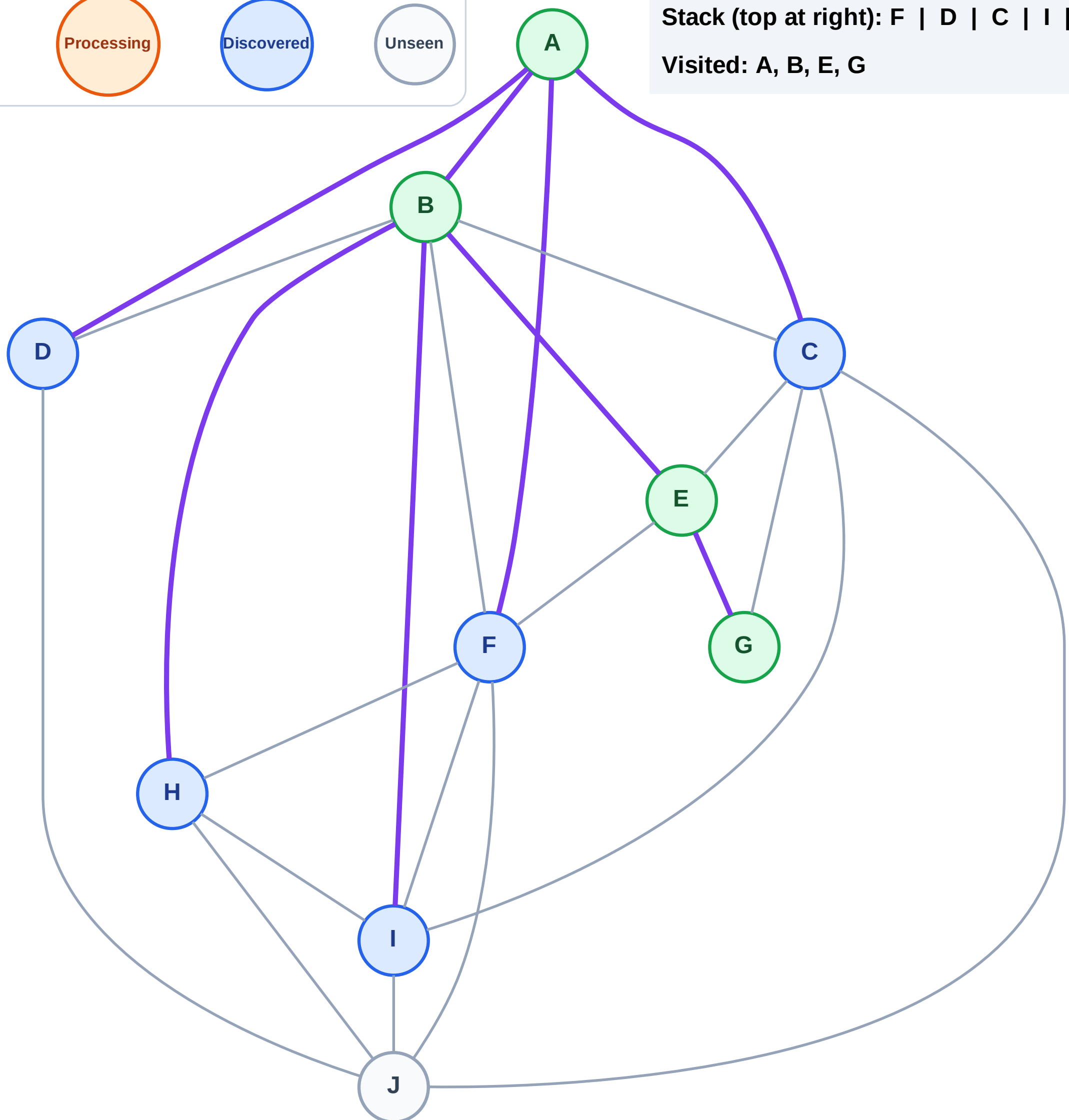
Discovered



Unseen

Stack (top at right): F | D | C | I | H

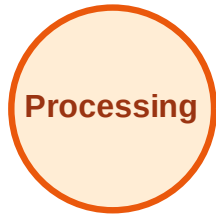
Visited: A, B, E, G



DFS Traversal

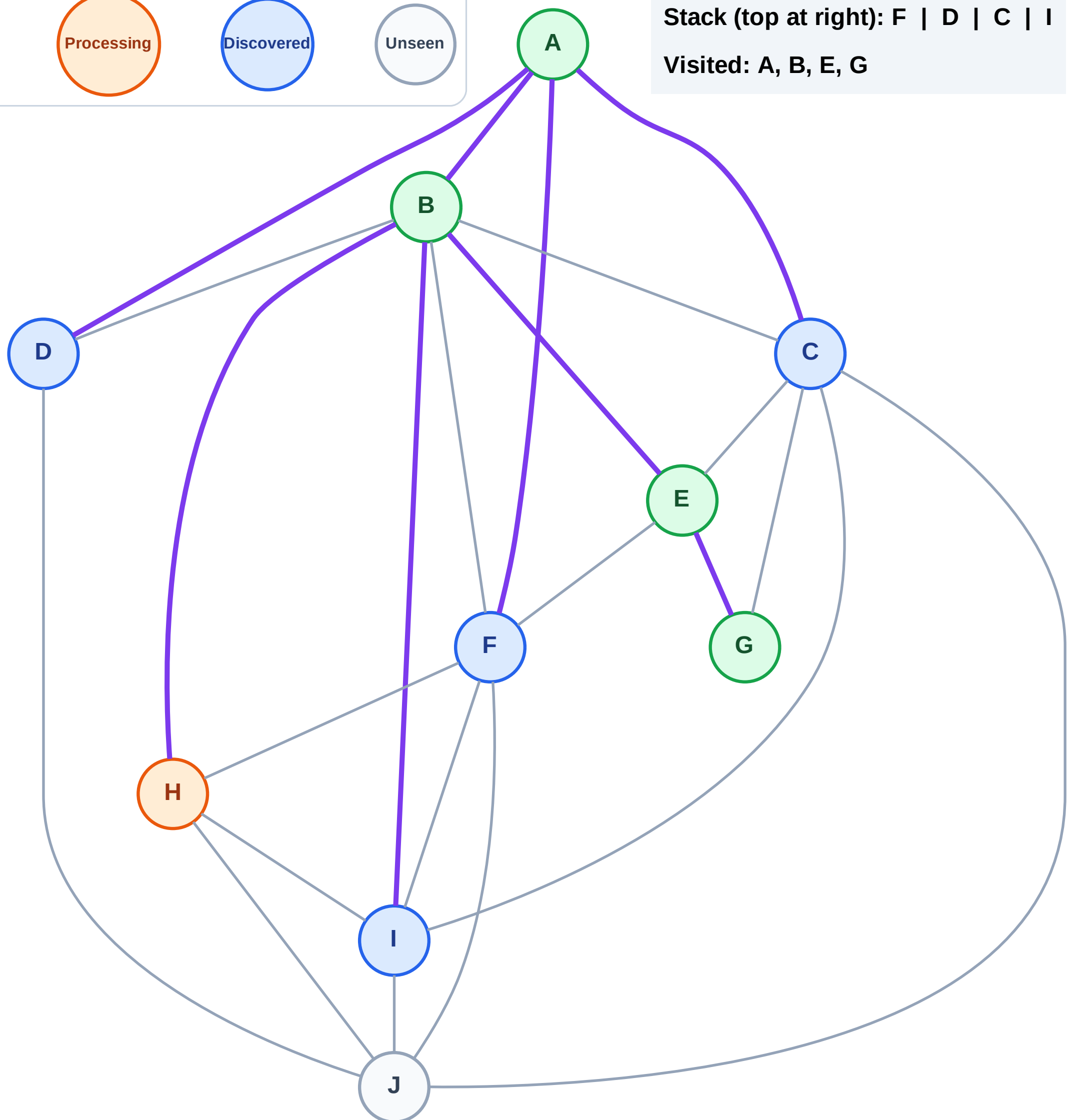
Step 10: Process node H

Legend



Stack (top at right): F | D | C | I

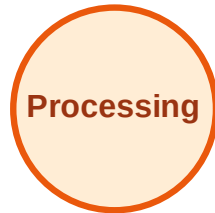
Visited: A, B, E, G



DFS Traversal

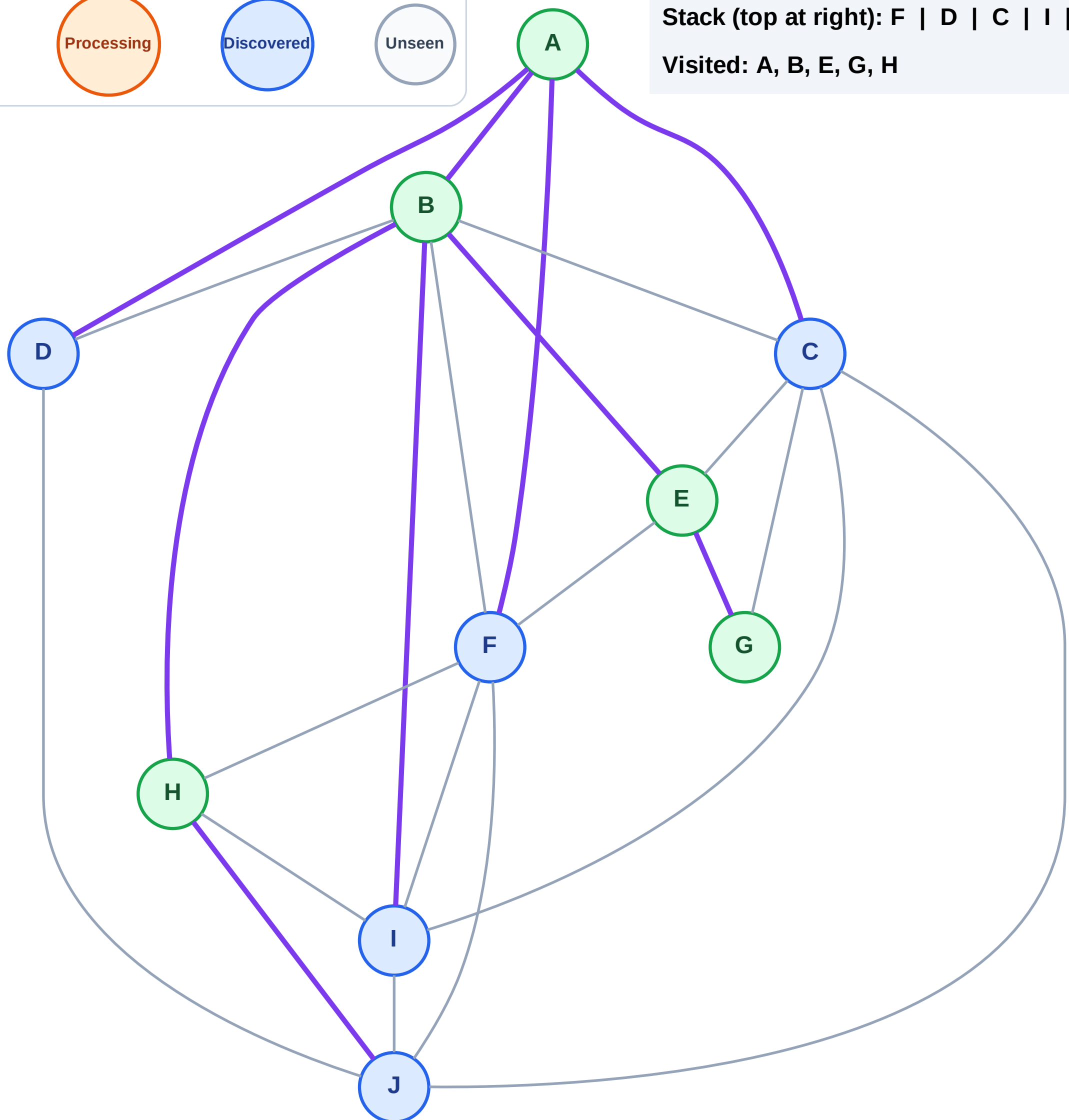
Step 11: Discover J from H

Legend



Stack (top at right): F | D | C | I | J

Visited: A, B, E, G, H



DFS Traversal

Step 12: Process node J

Legend

Complete

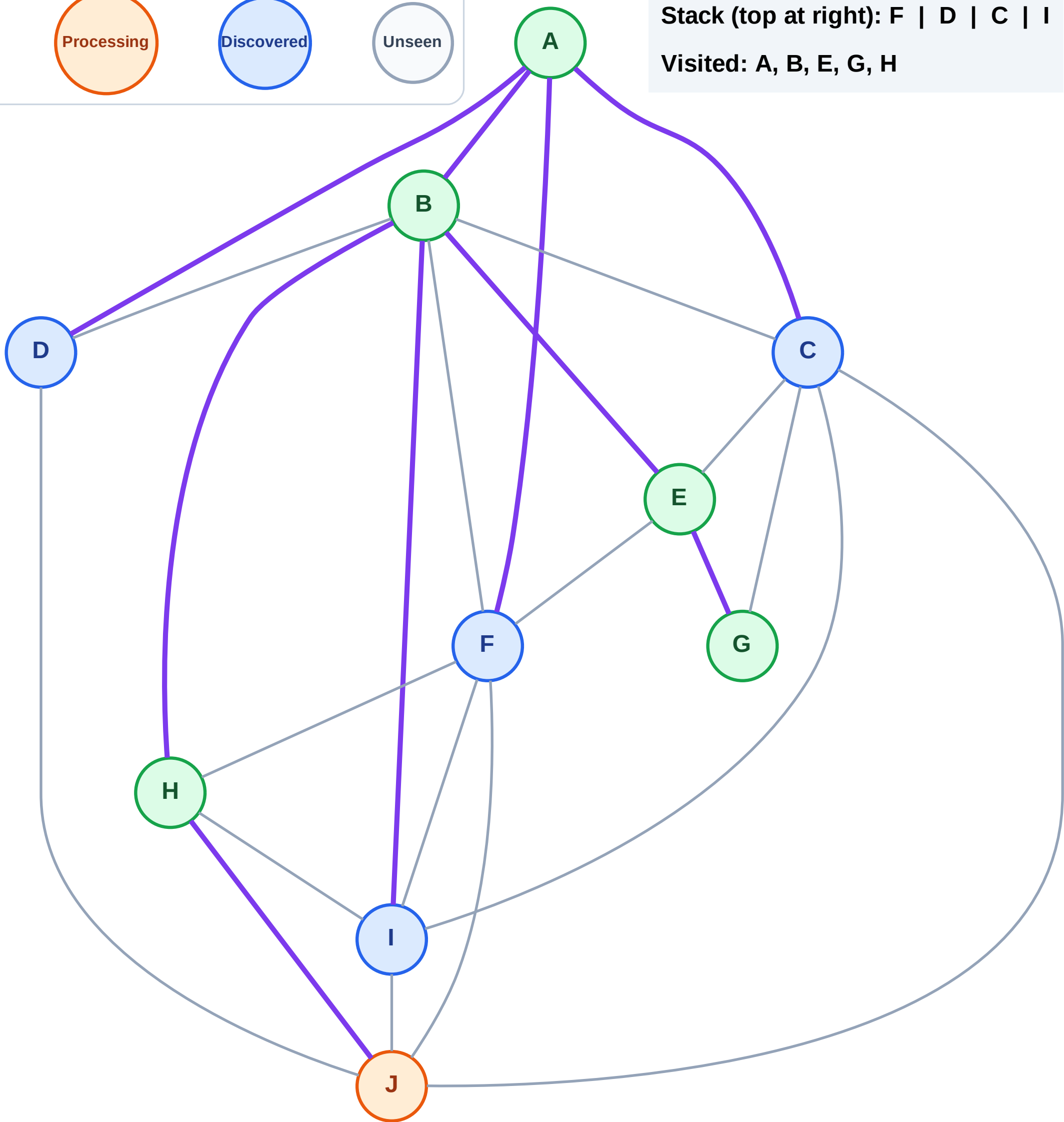
Processing

Discovered

Unseen

Stack (top at right): F | D | C | I

Visited: A, B, E, G, H



DFS Traversal

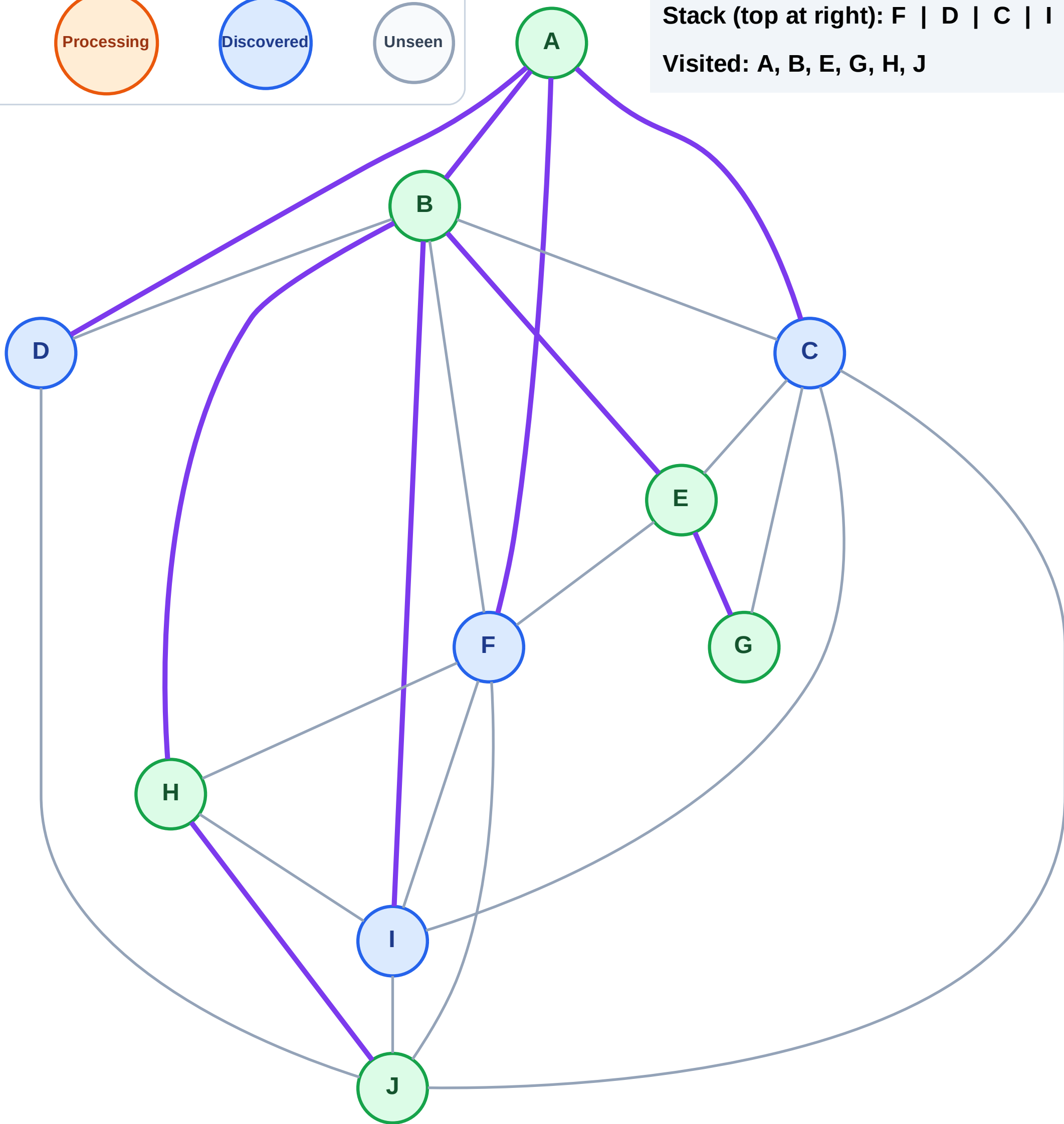
Step 13: No new neighbors from J

Legend



Stack (top at right): F | D | C | I

Visited: A, B, E, G, H, J

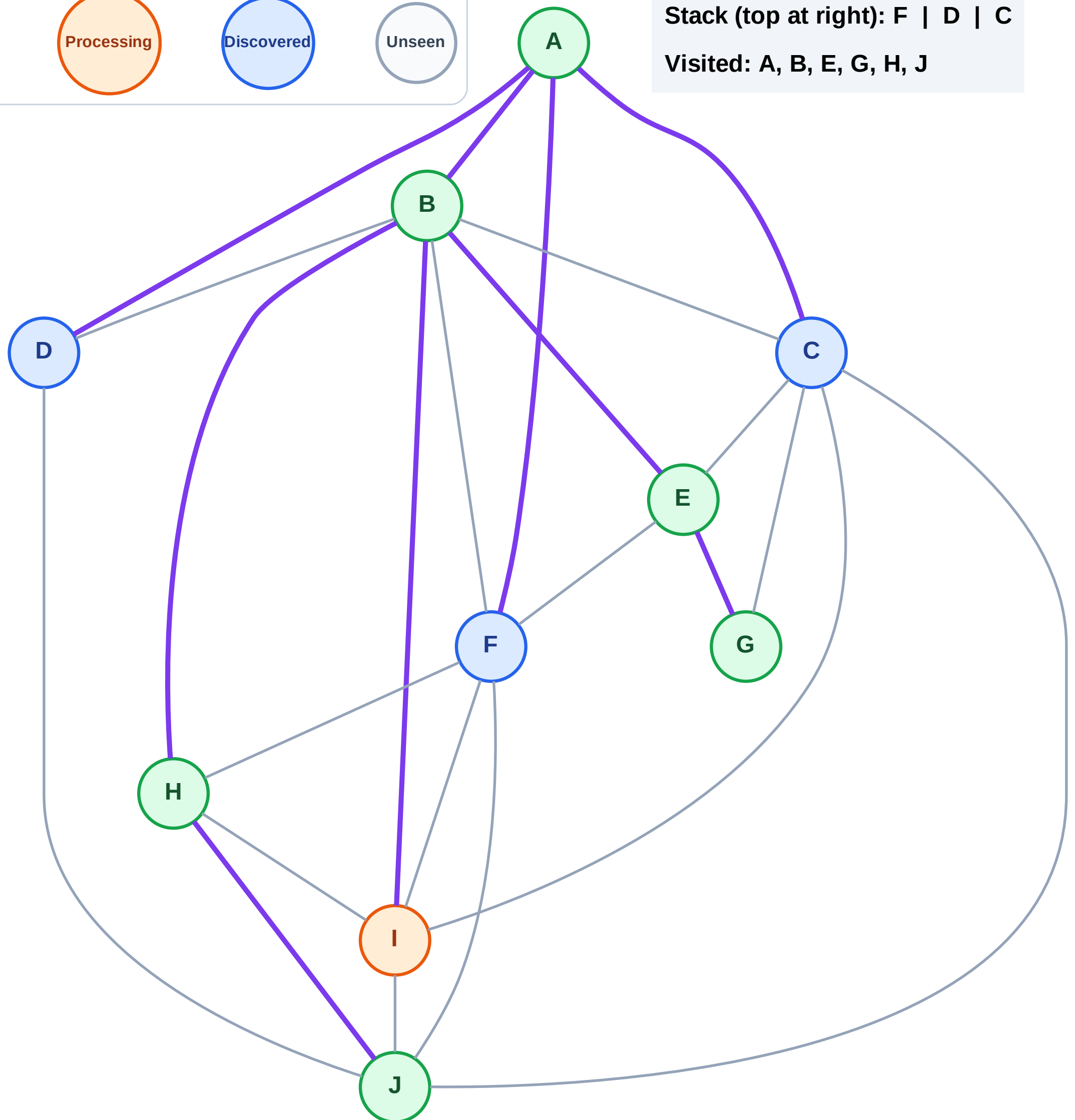


Step 14: Process node I

Page 10 of 10



Visited: A B E G H I



DFS Traversal

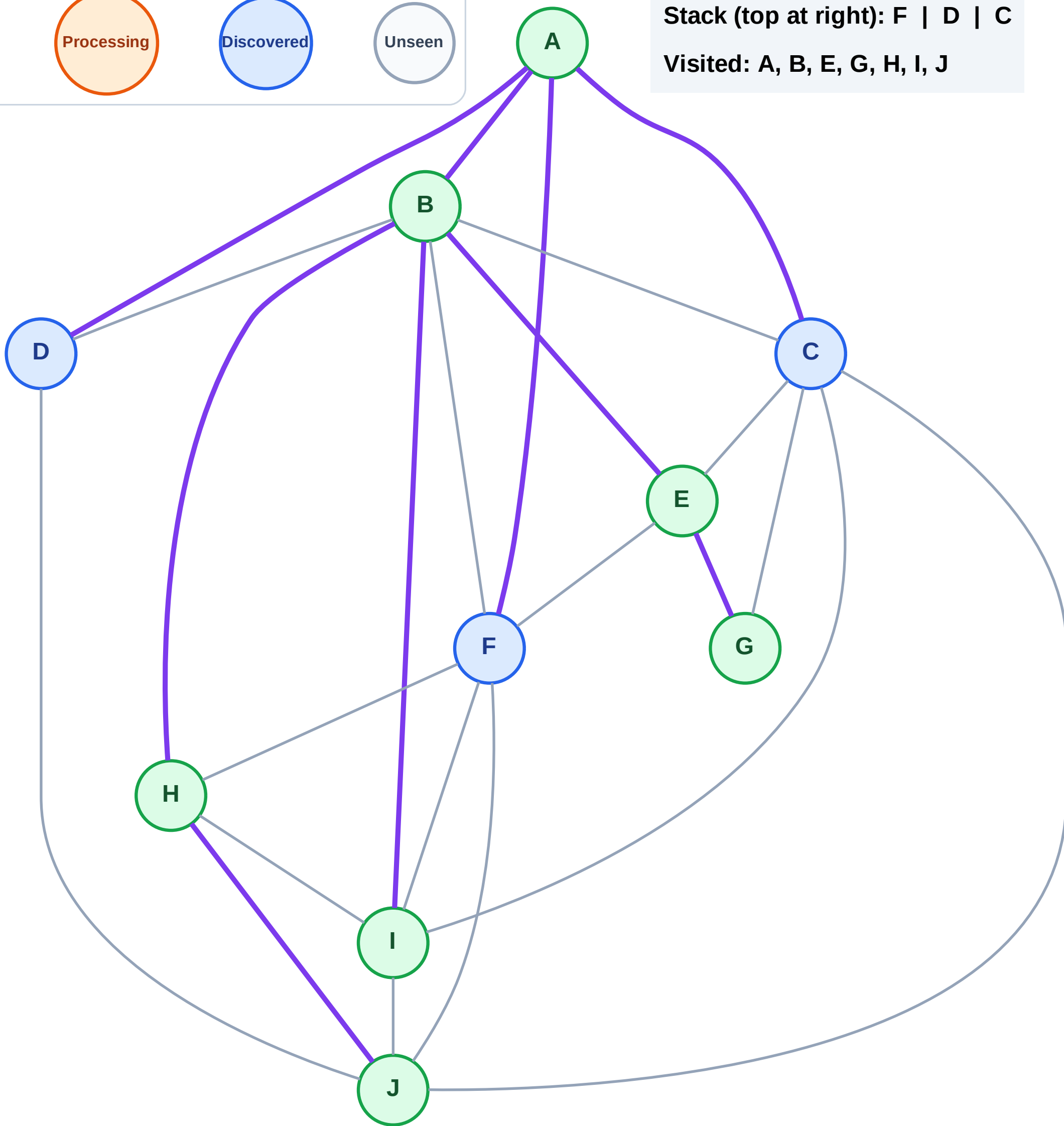
Step 15: No new neighbors from I

Legend



Stack (top at right): F | D | C

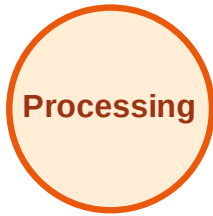
Visited: A, B, E, G, H, I, J



DFS Traversal

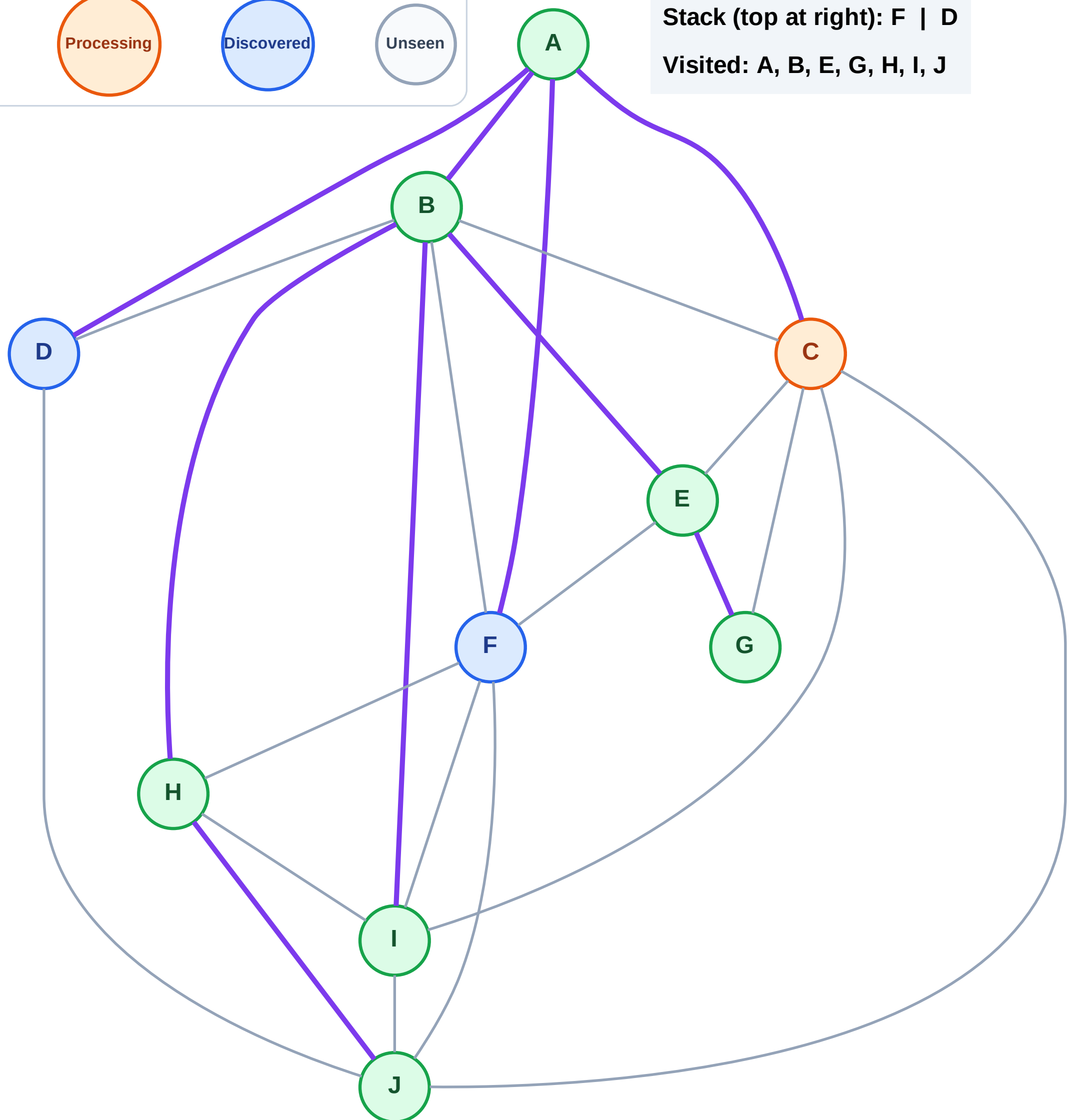
Step 16: Process node C

Legend



Stack (top at right): F | D

Visited: A, B, E, G, H, I, J



DFS Traversal

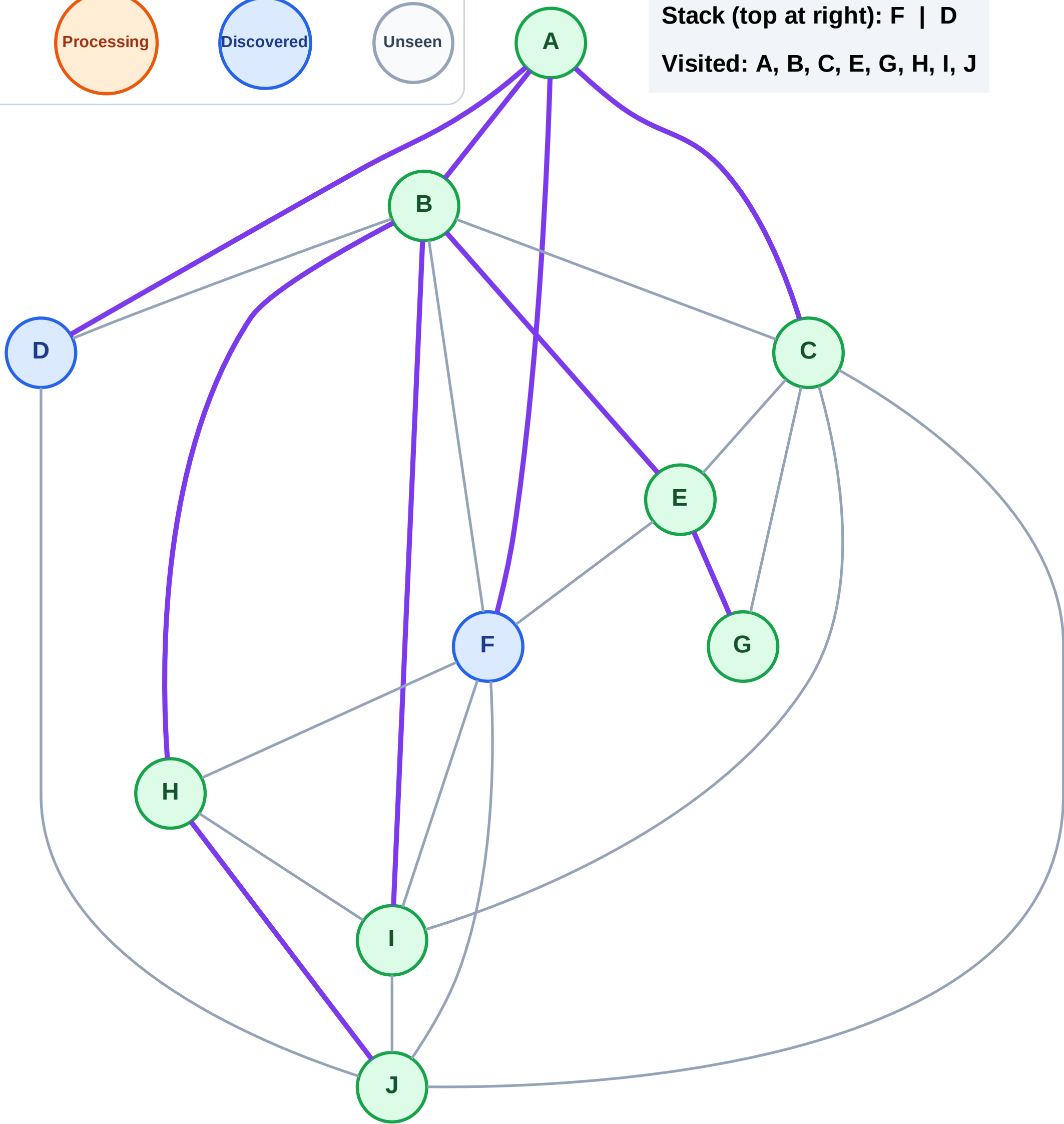
Step 17: No new neighbors from C

Legend



Stack (top at right): F | D

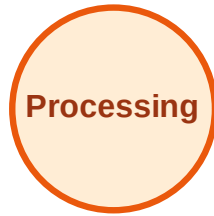
Visited: A, B, C, E, G, H, I, J



DFS Traversal

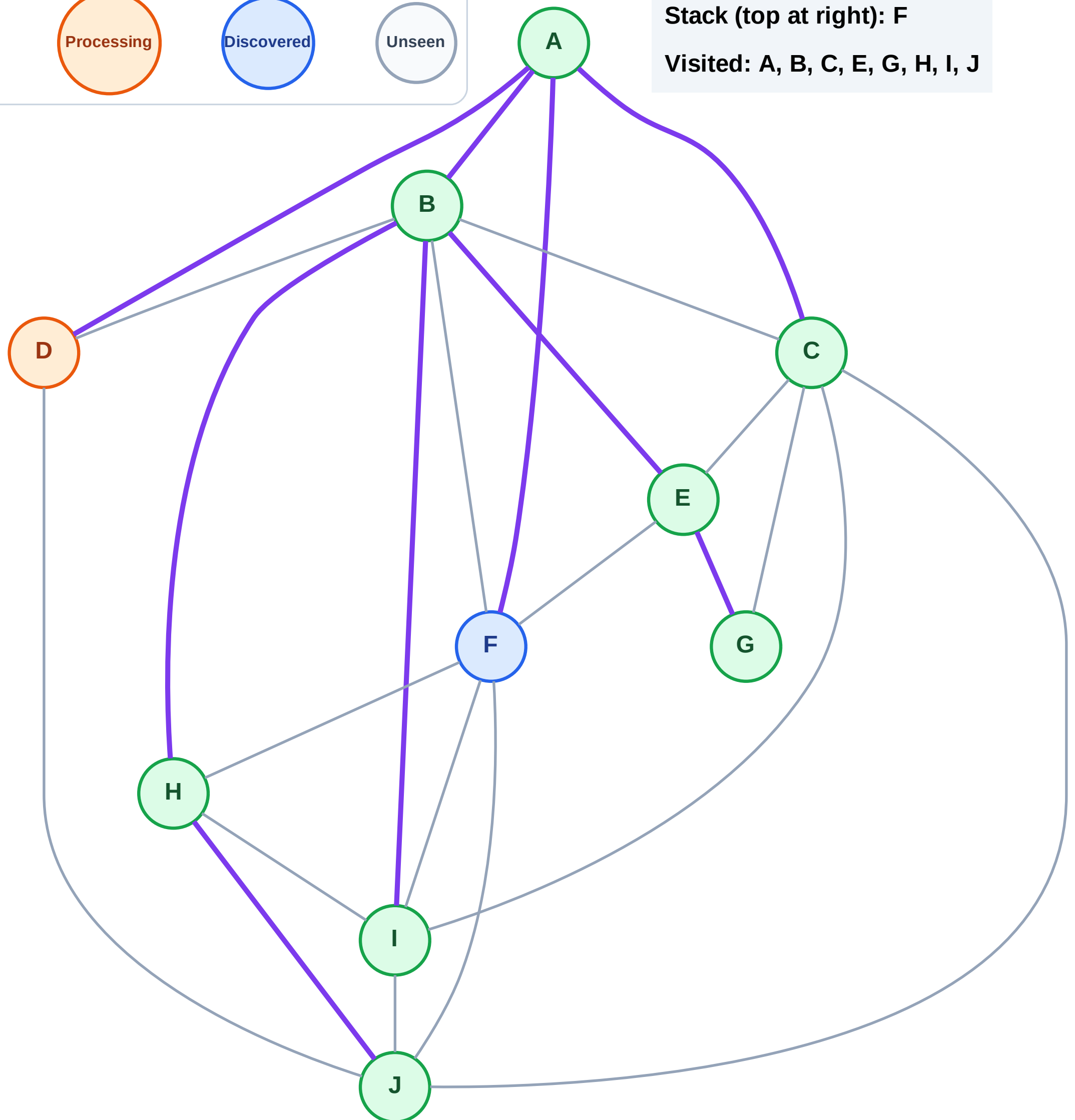
Step 18: Process node D

Legend



Stack (top at right): F

Visited: A, B, C, E, G, H, I, J



DFS Traversal

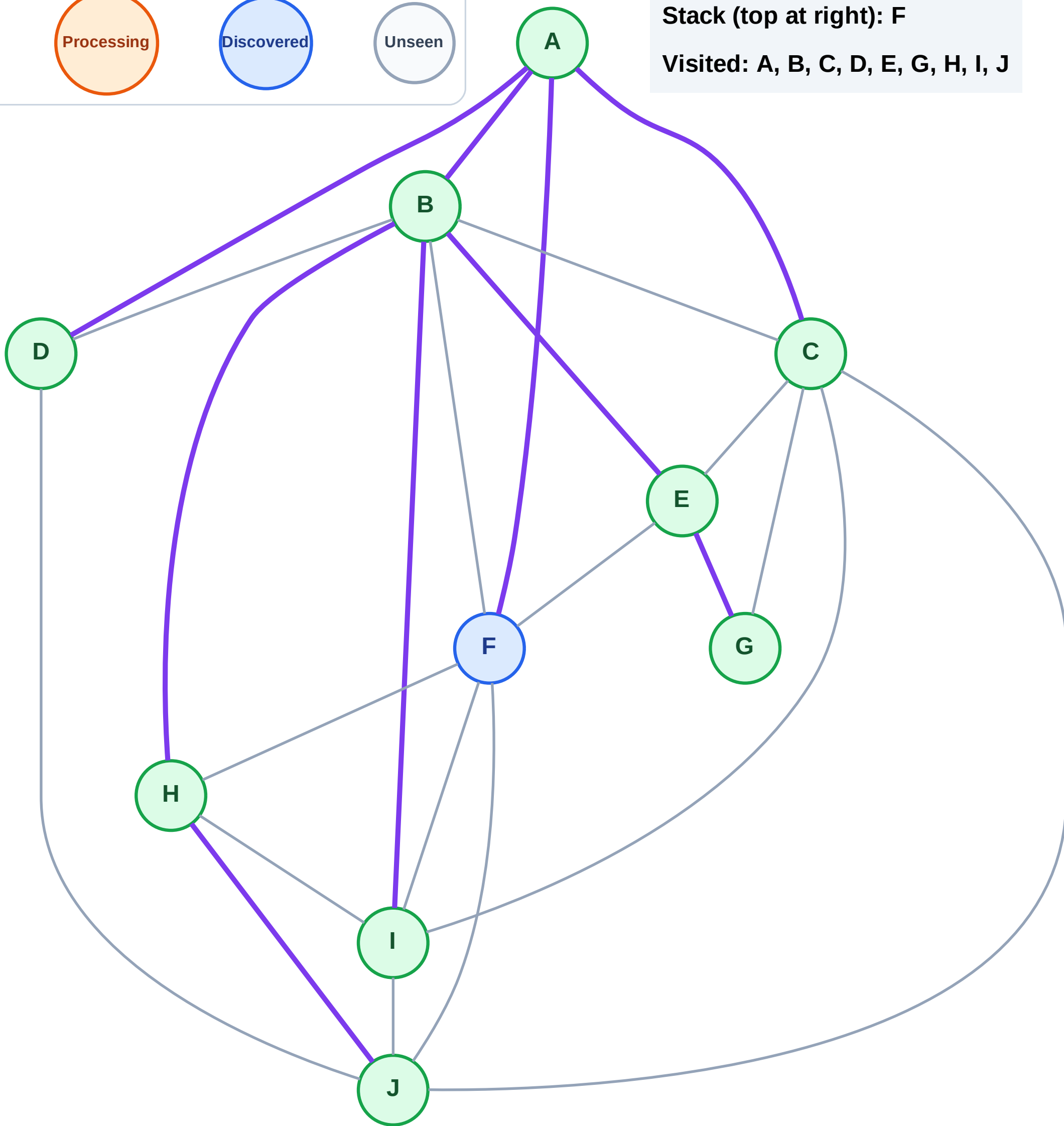
Step 19: No new neighbors from D

Legend



Stack (top at right): F

Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

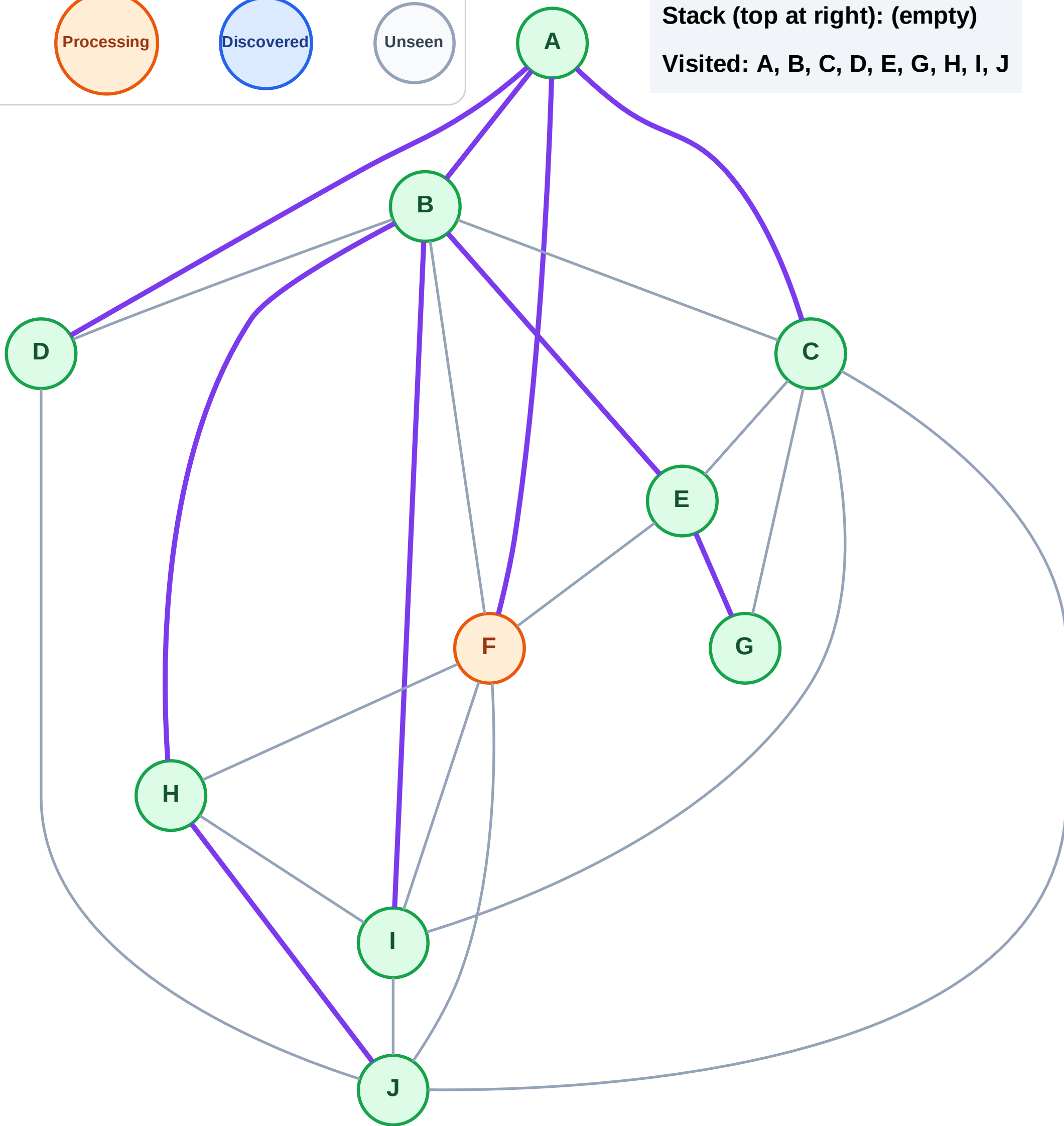
Step 20: Process node F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

